Working Paper Report of Moral Judgement under Two-sided Tobit Models

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This run uses `Version 2.0/consolidado_ALL_2026_04_09_LONG.xlsx` as the only analytical source and preserves each imported row as one real judgement observation. The production estimator is a two-sided Tobit fitted with `survival::survreg`, using bilateral censoring at `-9` and `9`, participant-cluster robust standard errors through `cluster = id`, and `factor(session)` in every active formula.

1 Understanding the modeling strategy

The purpose of the pipeline is to explain how participants assign moral judgement to a focal negotiator within a structured experimental setting. The objective is not limited to describing average responses. Rather, the pipeline is designed to estimate how judgement changes as a function of empathy, group alignment, negotiation decisions, and role-specific relational structure. The outcome of interest is `judgement`, interpreted as the participant's moral evaluation of the target negotiator. Because each participant contributes repeated evaluations across scenarios and targets, the modeling strategy must satisfy three conditions simultaneously: it must preserve one row per real observation, respect the bounded structure of the outcome, and account for within-participant dependence. For these reasons, the production workflow uses a two-sided Tobit model with participant-cluster robust inference and session adjustment.

The need for a Tobit specification follows directly from the nature of the dependent variable. `judgement` is continuous in interpretation but bounded in measurement. Values at the extremes are therefore not ordinary unrestricted observations; they are limits imposed by the response scale. A standard linear model assumes an outcome that can vary freely over the real line, which is inappropriate when responses accumulate at lower and upper boundaries. The Tobit framework addresses this by assuming an underlying latent moral evaluation, denoted here by `y_i^*`, that is only observed through a bounded realized score:

$$y_i^* = X_i\beta + \varepsilon_i$$

and

$$judgement_i = \max(-9, \min(9, y_i^*)).$$

Under this specification, the model jointly represents three kinds of outcomes: interior continuous values, lower-bound pile-up, and upper-bound pile-up. In this context, censoring does not mean that data are missing. It means that the observed score is recorded at the measurement boundary whenever the latent evaluation would extend beyond the scale. The Tobit model is therefore not a technical convenience but a direct response to the design of the judgement variable itself.

A second methodological challenge arises from the repeated-measures structure of the experiment. Each participant contributes multiple rows, so the observations are not independent. Ignoring this dependence would tend to underestimate standard errors and overstate statistical significance. The current pipeline addresses this issue through participant-cluster robust standard errors using `id` as the clustering variable. Session is also incorporated because different experimental sessions may vary in context, timing, or implementation details. This is handled through `factor(session)`, which absorbs systematic session-level shifts. In the active production branch, session is therefore modeled as a fixed-effect adjustment rather than as a random session intercept. The implemented estimator is best described as a two-sided Tobit with participant-cluster robust inference and session fixed effects.

The predictors are organized into theoretically meaningful blocks. The first block contains empathy dimensions, represented by `iri_fs`, `iri_ec`, `iri_pt`, and `iri_pd`, and provides the basis for H1. The second block captures role-specific ingroup/outgroup structure and underlies H2. The third block incorporates negotiation decisions through `decision_target`, `decision_other`, and their interaction, forming the core of H4. Finally, all models include sociodemographic controls such as age, SES, sex, and player faculty. This structure allows the pipeline to evaluate distinct explanatory mechanisms while preserving a coherent hypothesis architecture.

Interactions are especially important in this design because additive effects alone are unlikely to capture the logic of the experiment. In H3, empathy-by-group interactions test whether the effect of empathy depends on social alignment. In H4 and H5, the interaction between `decision_target` and `decision_other` evaluates whether the moral meaning of one negotiator's decision depends on the counterpart's decision. These interaction terms are theoretically motivated and reflect the fact that moral evaluation is shaped not only by isolated attributes but also by combinations of dispositions, relationships, and actions.

The distinction between victim and bystander roles is central to the entire modeling strategy. Victim models focus on victim-negotiator alignment and therefore operate with a more direct relational structure. Bystander models, by contrast, require a broader map that includes bystander-victim, bystander-negotiator, and victim-negotiator relations. Because these mechanisms differ substantively, the victim and bystander specifications cannot be treated as interchangeable. Role-specific models are therefore necessary both statistically and theoretically.

The five hypothesis families follow naturally from this structure. H1 models `judgement` as a function of empathy and controls. H2 focuses on role-specific group alignment. H3 combines empathy and group structure with theoretically motivated interactions. H4 models judgement as a function of `decision_target`, `decision_other`, and their interaction. H5 integrates all previous components into a single specification. This progression allows the analysis to move from simpler explanations toward a more comprehensive account of moral judgement.

A key principle of the workflow is that one row remains one real target-judgement observation. The pipeline does not duplicate rows into separate pseudo-observations for negotiators, because doing so would artificially inflate the sample size and distort inference. Instead, the relational context involving N1, N2, victim, and bystander is reconstructed within each existing row. This preserves the integrity of the long-format design while maintaining the proper unit of analysis.

Taken together, the current production strategy has several strengths. It respects the bounded structure of the outcome, preserves the long-format observational design, adjusts inference for repeated observations, controls for session-level heterogeneity, separates victim and bystander mechanisms, and maps directly onto the theoretical architecture of H1 through H5. At the same time, its limitations must also be recognized. The estimator is not a full mixed-effects Tobit with random participant and session intercepts. In addition, sparse role-specific cells may still produce rank-deficient contrasts in interaction-heavy models. For that reason, interpretation should rely on the full combination of coefficient tables and model-implied figures rather than on isolated p-values.

Overall, the current production pipeline is statistically and conceptually aligned with the experiment. It treats bounded continuous moral judgement with a two-sided Tobit model, handles repeated observations through participant-cluster robust inference, adjusts session heterogeneity through fixed effects, and preserves role-specific relational theory within model specification. This makes the present workflow a defensible production framework, while also leaving a clear path for future multilevel Tobit extensions.

2 Semantic naming bridge and row-level mapping

Legacy/intuitive naming bridge: `accept_target` -> active operational name `decision_target`; `accept_other` -> active operational name `decision_other`.

Row-level semantics: `target` and `other` are dynamic roles per observation, while `N1` and `N2` are structural slots reconstructed inside each row.

Table 1. Row-level decision mapping from dynamic target/other to structural N1/N2 slots

rule	expected_mapping	rows_in_scope	rows_following_rule	status
target == 1	N1_decision = decision_target; N2_decision = decision_other	2430	2430	PASS
target == 2	N1_decision = decision_other; N2_decision = decision_target	2430	2430	PASS

3 Dataset and sample description

The report uses the consolidated long experimental dataset as the single analytical source. Each participant contributes 20 judgement rows in principle: ten scenarios multiplied by two target-negotiator evaluations. Each imported row remains one real judgement observation on the target negotiator, enriched with relational context for N1, N2, victim, and bystander without duplicating rows. Double counting is prevented because N1 and N2 are reconstructed as contextual attributes inside each existing row rather than by expanding the file into duplicated negotiator-specific observations. Victim and bystander analyses are estimated separately so that relational coding follows the role-specific logic of the experiment.

The authoritative interpretation is that each player observes ten scenarios and evaluates two negotiators, so the longitudinal file should contain 20 judgement rows per participant. The clustering diagnostic below is consistent with that design.

Table 2. Participant summary

metric	value
participants	243.000
sessions	16.000
mean_age	20.086

metric	value
women_share	0.426
engineering_share	0.568

Table 3. Judgement summary

sample	rows	participants	mean_judgement	sd_judgement
All	4860	243	1.623	6.674
Victim	2430	243	1.504	6.842
Bystander	2430	243	1.741	6.500

4 Datacard and symbol dictionary

Table 4. Datacard symbol dictionary

symbol	definition
judgement	Observed moral judgement on the bounded scale from -9 to 9.
y*	Latent judgement tendency underlying the censored Tobit observation.
iri_fs / iri_ec / iri_pt / iri_pd	IRI empathy dimensions: fantasy, empathic concern, perspective taking, and personal distress.
target (row-dynamic)	Judged negotiator in that row; this role is dynamic and can be N1 or N2 depending on the observation.
other (row-dynamic counterpart)	Counterpart negotiator in that same row context (the non-target actor).
N1 / N2 (structural slots)	Structural negotiator identities reconstructed within each row for relational modeling; they are not fixed aliases of target/other.
decision_target	Indicator for whether the row-dynamic target negotiator accepted the harmful deal; active operational name for legacy wording accept_target.
decision_other	Indicator for whether the row-dynamic other negotiator accepted the harmful deal; active operational name for legacy wording accept_other.

symbol	definition
victim_N1_group / victim_N2_group	Victim-specific relations to negotiator 1 and negotiator 2, with ingroup defined by faculty coincidence including control-control matches.
bystander_victim_group / bystander_N1_group / bystander_N2_group	Bystander-side relational factors for the victim and both negotiators, again using faculty coincidence as ingroup.
group_target / group_other (legacy audit)	Legacy source grouping fields retained for provenance checks; not used directly in active H2/H3/H5 formulas.
N1_N2_same_faculty	Context term indicating whether N1 and N2 share faculty membership.
factor(session)	Session fixed effects included directly in every fitted formula.
cluster = id	Participant-level clustering used for robust standard errors and repeated-measures adjustment.
Log(scale)	Estimated Tobit log-scale parameter summarizing latent residual dispersion.

Table 5. Observation audit

checkpoint	value
base_excel_rows	4860
processed_import_rows	4860
processed_judgment_rows	4860
unique_source_row_numbers	4860
duplicated_source_row_numbers	0

5 Predictor glossary

Table 6. Predictor glossary (reader version)

Code	Interpretation
FS	Fantasy empathy dimension.
EC	Empathic concern empathy dimension.
PT	Perspective-taking empathy dimension.
PD	Personal-distress empathy dimension.

Code	Interpretation
V-N1 In	Victim and N1 are from the same faculty.
V-N1 Out	Victim and N1 are from different faculties.
V-N2 In	Victim and N2 are from the same faculty.
V-N2 Out	Victim and N2 are from different faculties.
B-V Out	Bystander and victim are from different faculties.
B-N1 In	Bystander and N1 are from the same faculty.
B-N1 Out	Bystander and N1 are from different faculties.
B-N2 In	Bystander and N2 are from the same faculty.
B-N2 Out	Bystander and N2 are from different faculties.
SameFac	N1 and N2 share faculty membership.
FS x V-N1 Out	Fantasy slope difference when victim-N1 is outgroup rather than ingroup.
EC x V-N2 Out	Empathic-concern slope difference when victim-N2 is outgroup rather than ingroup.
PT x B-V Out	Perspective-taking slope difference when the bystander-victim relation is outgroup rather than ingroup.
PD x B-N1 Out	Personal-distress slope difference when the bystander-N1 relation is outgroup rather than ingroup.
Target Acc	Row-dynamic target negotiator accepted the harmful deal (legacy wording: accept_target).
Other Acc	Row-dynamic counterpart negotiator accepted the harmful deal (legacy wording: accept_other).
Target x Other	Joint decision effect when both negotiator decisions are considered together.
Eng part.	Participant belongs to Engineering, relative to Humanities.
Woman	Participant is a woman.
Age	Participant age.

Code	Interpretation
SES	Participant socioeconomic status.

Note. Group contrasts are interpreted against the ingroup baseline unless explicitly stated otherwise.

The report keeps compact predictor references in figure captions and narratives, but the glossary above remains the authoritative mapping back to the current pipeline variables.

6 Interaction interpretation rules

1. When an interaction is statistically relevant, the main effects should be read as the baseline component of the relationship rather than the whole substantive story.
2. Continuous-by-factor interactions indicate that the empathy slope changes across relational conditions.
3. Factor-by-factor interactions indicate that the joint context differs from what would be expected by adding the two main contrasts independently.
4. The target-by-other decision interaction indicates that the moral meaning of one negotiator's choice depends on what the counterpart did.
5. Session effects are adjustment terms only and are not interpreted as substantive experimental mechanisms.

7 H1-H5 hypotheses with role-specific equation summaries

Table 1: H1-H5 role-specific formulas and theoretical focus.

H	Role	Formula	Focus
H1	Victim	iri__fs + iri__ec + iri__pt + iri__pd + age + ses + sex__female + faculty__player__factor + factor(session)	Empathy dimensions only, always adjusted by sociodemographics.
H1	Bystander	iri__fs + iri__ec + iri__pt + iri__pd + age + ses + sex__female + faculty__player__factor + factor(session)	Empathy dimensions only, always adjusted by sociodemographics.
H2	Victim	victim__N1__group + victim__N2__group + victim__N1__group : victim__N2__group + N1__N2__same__faculty + age + ses + sex__female + faculty__player__factor + factor(session)	Victim-side ingroup/outgroup structure with the allowed N1 x N2 relational interaction.
H2	Bystander	bystander__victim__group + bystander__N1__group + bystander__N2__group + victim__N1__group + victim__N2__group + bystander__N1__group : bystander__N2__group + victim__N1__group : victim__N2__group + N1__N2__same__faculty + age + ses + sex__female + faculty__player__factor + factor(session)	Bystander-side relational structure with explicit bystander-victim, bystander-negotiator, victim-negotiator, and N1/N2 context terms.

H	Role	Formula	Focus
H3	Victim	iri__fs + iri__ec + iri__pt + iri__pd + victim__N1__group + victim__N2__group + victim__N1__group : victim__N2__group + N1__N2__same__faculty + iri__fs : victim__N1__group + iri__fs : victim__N2__group + iri__ec : victim__N1__group + iri__ec : victim__N2__group + iri__pt : victim__N1__group + iri__pt : victim__N2__group + iri__pd : victim__N1__group + iri__pd : victim__N2__group + age + ses + sex__female + faculty__player__factor + factor(session)	Empathy plus victim-side relational structure, including empathy x victim-N1 and empathy x victim-N2 interactions because empathy may depend on negotiator closeness.
H3	Bystander	iri__fs + iri__ec + iri__pt + iri__pd + bystander__victim__group + bystander__N1__group + bystander__N2__group + victim__N1__group + victim__N2__group + bystander__N1__group : bystander__N2__group + victim__N1__group : victim__N2__group + N1__N2__same__faculty + iri__fs : bystander__victim__group + iri__fs : bystander__N1__group + iri__fs : bystander__N2__group + iri__ec : bystander__victim__group + iri__ec : bystander__N1__group + iri__ec : bystander__N2__group + iri__pt : bystander__victim__group + iri__pt : bystander__N1__group + iri__pt : bystander__N2__group + iri__pd : bystander__victim__group + iri__pd : bystander__N1__group + iri__pd : bystander__N2__group + age + ses + sex__female + faculty__player__factor + factor(session)	Empathy plus bystander-side relational structure, including empathy x bystander-victim and empathy x bystander-negotiator interactions because empathy may depend on group closeness in the bystander role.
H4	Victim	decision__target * decision__other + age + ses + sex__female + faculty__player__factor + factor(session)	Target and other negotiator decisions with their interaction, plus sociodemographics.
H4	Bystander	decision__target * decision__other + age + ses + sex__female + faculty__player__factor + factor(session)	Target and other negotiator decisions with their interaction, plus sociodemographics.
H5	Victim	iri__fs + iri__ec + iri__pt + iri__pd + victim__N1__group + victim__N2__group + victim__N1__group : victim__N2__group + N1__N2__same__faculty + iri__fs : victim__N1__group + iri__fs : victim__N2__group + iri__ec : victim__N1__group + iri__ec : victim__N2__group + iri__pt : victim__N1__group + iri__pt : victim__N2__group + iri__pd : victim__N1__group + iri__pd : victim__N2__group + decision__target * decision__other + age + ses + sex__female + faculty__player__factor + factor(session)	Integrated model with empathy, victim-side relations, empathy x group interactions, decisions, and the victim-side relational interaction.
H5	Bystander	iri__fs + iri__ec + iri__pt + iri__pd + bystander__victim__group + bystander__N1__group + bystander__N2__group + victim__N1__group + victim__N2__group + bystander__N1__group : bystander__N2__group + victim__N1__group : victim__N2__group + N1__N2__same__faculty + iri__fs : bystander__victim__group + iri__fs : bystander__N1__group + iri__fs : bystander__N2__group + iri__ec : bystander__victim__group + iri__ec : bystander__N1__group + iri__ec : bystander__N2__group + iri__pt : bystander__victim__group + iri__pt : bystander__N1__group + iri__pt : bystander__N2__group + iri__pd : bystander__victim__group + iri__pd : bystander__N1__group + iri__pd : bystander__N2__group + decision__target * decision__other + age + ses + sex__female + faculty__player__factor + factor(session)	Integrated model with empathy, bystander-side relations, empathy x group interactions, decisions, and the role-specific relational interactions.

Any earlier repository note that treated negotiator code `0` as anything other than the explicit control category, or that narrowed H3 to additive effects only, should now be treated as outdated. The active formulas below are the authoritative specification.

7.1 H1

Victim: `iri_fs + iri_ec + iri_pt + iri_pd + age + ses + sex_female + faculty_player_factor + factor(session)`

Bystander: `iri_fs + iri_ec + iri_pt + iri_pd + age + ses + sex_female + faculty_player_factor + factor(session)`

7.2 H2

Victim: `victim_N1_group + victim_N2_group + victim_N1_group:victim_N2_group + N1_N2_same_faculty + age + ses + sex_female + faculty_player_factor + factor(session)`

Bystander: `bystander_victim_group + bystander_N1_group + bystander_N2_group + victim_N1_group + victim_N2_group + bystander_N1_group:bystander_N2_group + victim_N1_group:victim_N2_group + N1_N2_same_faculty + age + ses + sex_female + faculty_player_factor + factor(session)`

7.3 H3

Victim: `iri_fs + iri_ec + iri_pt + iri_pd + victim_N1_group + victim_N2_group + victim_N1_group:victim_N2_group + N1_N2_same_faculty + iri_fs:victim_N1_group + iri_fs:victim_N2_group + iri_ec:victim_N1_group + iri_ec:victim_N2_group + iri_pt:victim_N1_group + iri_pt:victim_N2_group + iri_pd:victim_N1_group + iri_pd:victim_N2_group + age + ses + sex_female + faculty_player_factor + factor(session)`

Bystander: `iri_fs + iri_ec + iri_pt + iri_pd + bystander_victim_group + bystander_N1_group + bystander_N2_group + victim_N1_group + victim_N2_group + bystander_N1_group:bystander_N2_group + victim_N1_group:victim_N2_group + N1_N2_same_faculty + iri_fs:bystander_victim_group + iri_fs:bystander_N1_group + iri_fs:bystander_N2_group + iri_ec:bystander_victim_group + iri_ec:bystander_N1_group + iri_ec:bystander_N2_group + iri_pt:bystander_victim_group + iri_pt:bystander_N1_group + iri_pt:bystander_N2_group + iri_pd:bystander_victim_group + iri_pd:bystander_N1_group + iri_pd:bystander_N2_group + age + ses + sex_female + faculty_player_factor + factor(session)`

7.4 H4

Victim: `decision_target * decision_other + age + ses + sex_female + faculty_player_factor + factor(session)`

Bystander: `decision_target * decision_other + age + ses + sex_female + faculty_player_factor + factor(session)`

7.5 H5

Victim: `iri_fs + iri_ec + iri_pt + iri_pd + victim_N1_group + victim_N2_group + victim_N1_group:victim_N2_group + N1_N2_same_faculty + iri_fs:victim_N1_group + iri_fs:victim_N2_group + iri_ec:victim_N1_group + iri_ec:victim_N2_group + iri_pt:victim_N1_group + iri_pt:victim_N2_group + iri_pd:victim_N1_group + iri_pd:victim_N2_group + decision_target * decision_other + age + ses + sex_female + faculty_player_factor + factor(session)`

Bystander: `iri_fs + iri_ec + iri_pt + iri_pd + bystander_victim_group + bystander_N1_group + bystander_N2_group + victim_N1_group + victim_N2_group + bystander_N1_group:bystander_N2_group + victim_N1_group:victim_N2_group + N1_N2_same_faculty + iri_fs:bystander_victim_group + iri_fs:bystander_N1_group + iri_fs:bystander_N2_group + iri_ec:bystander_victim_group + iri_ec:bystander_N1_group + iri_ec:bystander_N2_group + iri_pt:bystander_victim_group + iri_pt:bystander_N1_group + iri_pt:bystander_N2_group + iri_pd:bystander_victim_group + iri_pd:bystander_N1_group + iri_pd:bystander_N2_group + decision_target * decision_other + age + ses + sex_female + faculty_player_factor + factor(session)`

8 Mathematical foundations

The primary estimator is a two-sided Tobit fitted with `survival::survreg`.

$$y_i = \max(-9, \min(9, y_i^*))$$

$$y_i^* = \beta_0 + X_i\beta + \delta_{session(i)} + \varepsilon_i$$

where `factor(session)` supplies session fixed effects and cluster-robust standard errors are computed at the participant level through `cluster = id` with `robust = TRUE`. This report therefore treats session as an implemented fixed-effect adjustment, not as a random intercept.

In this production branch, `factor(session)` is reported instead of `(1|session)` because the fitted estimator is a two-sided Tobit with session fixed effects and participant-cluster robust standard errors. The report does not claim a random session intercept that was not actually estimated.

9 Dependence and effective sample size diagnostic

The following clustering diagnostic is descriptive. It summarizes within-participant dependence in the observed data and should not be read as evidence that the fitted estimator included participant random intercepts.

Table 7. Descriptive clustering diagnostic

metric	value
participants	243.000
observations	4860.000
average_observations_per_id	20.000
icc_descriptive	0.134
design_effect	3.537
effective_sample_size	1373.909

Because the target of inference is repeated judgement within participant, the effective-sample-size table is a descriptive clustering diagnostic only; it does not replace the model-based dependence adjustment through `cluster = id` and `factor(session)`.

10 Descriptive statistics and figures

Table 8. Decision summary by role

role_label	decision_pattern	n	mean_judgement
bystander	both_accept	604	-2.785
victim	both_accept	610	-3.167
bystander	both_reject	690	7.086
victim	both_reject	662	7.062
bystander	target_accept_other_reject	568	-3.155
victim	target_accept_other_reject	579	-3.803
bystander	target_reject_other_accept	568	4.958
victim	target_reject_other_accept	579	5.378

Table 9. Role-specific ingroup/outgroup summary

variable	level	n
victim_N1_group	ingroup	1682
victim_N1_group	outgroup	3178
victim_N2_group	ingroup	1578
victim_N2_group	outgroup	3282
bystander_victim_group	ingroup	1212
bystander_victim_group	outgroup	1218
bystander_N1_group	ingroup	812
bystander_N1_group	outgroup	1618

variable	level	n
bystander_N2_group	ingroup	798
bystander_N2_group	outgroup	1632
N1_N2_same_faculty	different	3276
N1_N2_same_faculty	same	1584

Table 10. Participant-level empathy and mean judgement correlation matrix

term	iri_fs	iri_ec	iri_pt	iri_pd	judgement
iri_fs	1.000	0.409	0.154	0.282	0.051
iri_ec	0.409	1.000	0.482	0.219	-0.017
iri_pt	0.154	0.482	1.000	-0.177	0.111
iri_pd	0.282	0.219	-0.177	1.000	-0.046
judgement	0.051	-0.017	0.111	-0.046	1.000

The group summary and the formulas above use role-specific ingroup/outgroup coding. Ingroup is defined by faculty coincidence, including `control` with `control`, while outgroup means non-matching faculties.

This figure summarizes the central empathy profile of the sample before conditioning on hypothesis-specific models.

These scatterplots show the participant-level descriptive relationship between empathy dimensions and average judgement.

This figure shows the raw shape of the bounded judgement outcome in the victim and bystander subsets.

This figure shows whether the raw bounded judgement distribution differs depending on whether the evaluated target is N1 or N2 within victim and bystander settings.

This figure isolates whether the target's own acceptance or rejection is associated with different raw judgement profiles within each role.

This figure shows whether judgement of the target varies with the acceptance or rejection of the other negotiator.

This figure shows how the target-focused judgement distribution changes across the four joint negotiation outcomes in victim and bystander settings.

This figure summarizes how often each joint decision pattern appears in victim and bystander subsets.

This figure shows how the raw judgement distribution varies across the full relational space defined by the faculty pairing of N1 and N2.

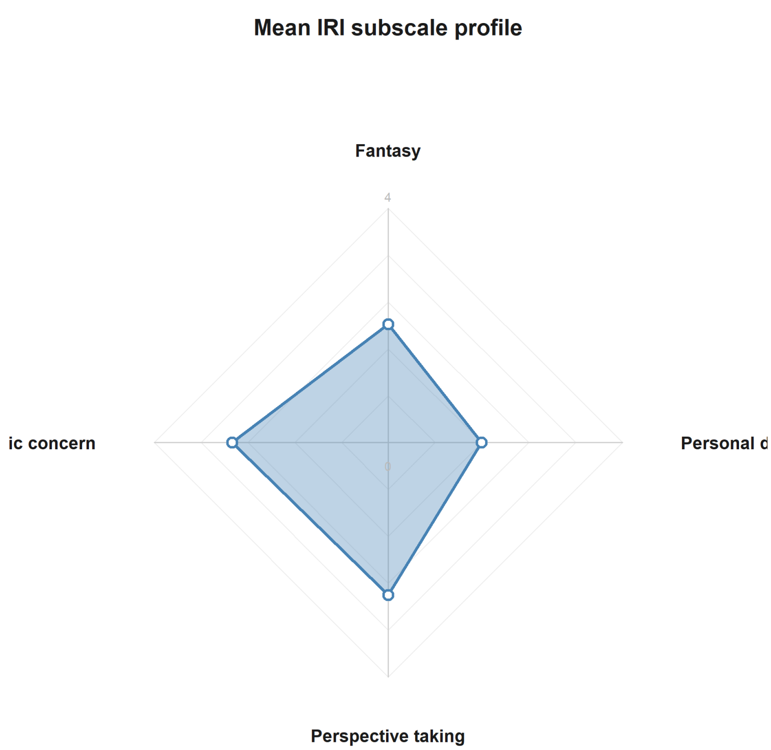

Figure 1: Mean IRI subscale profile across participants.

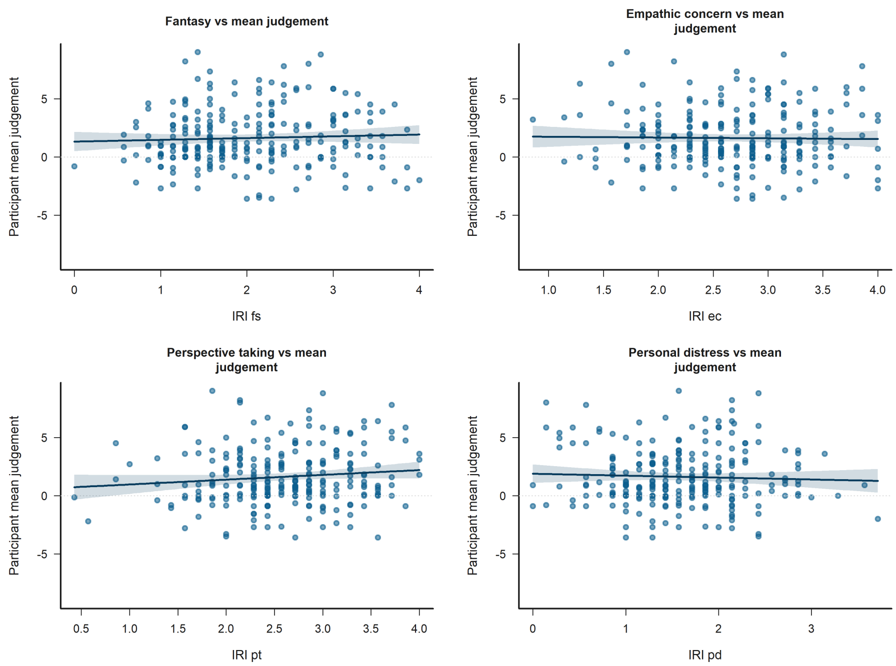

Figure 2: Participant-level bivariate scatters of IRI subscales against mean judgement.

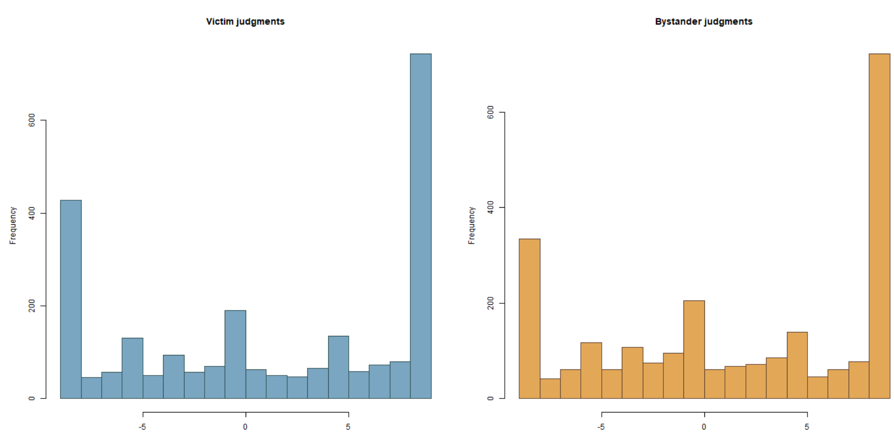

Figure 3: Observed judgement distributions split by role.

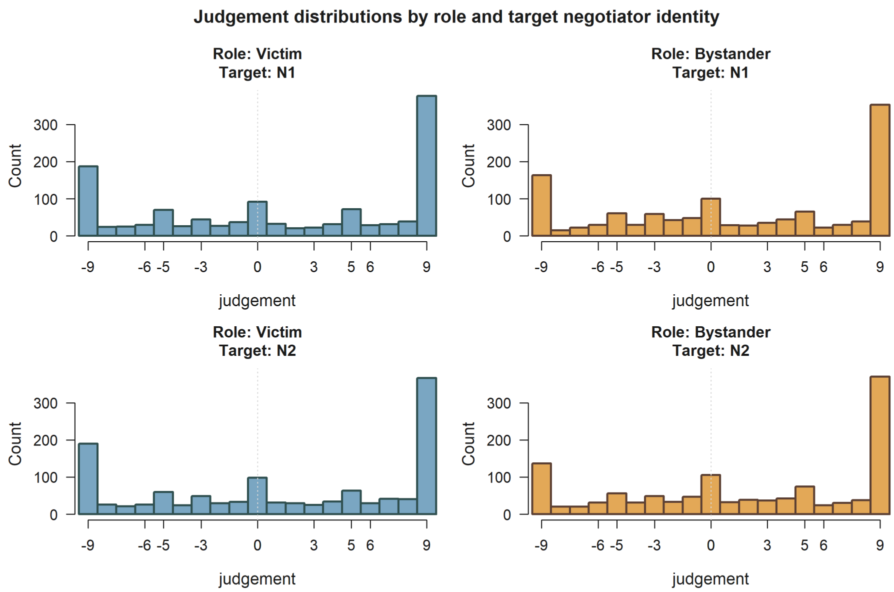

Figure 4: Observed judgement distributions by role and target negotiator.

Figure 5: Observed judgement distributions by role and target decision.

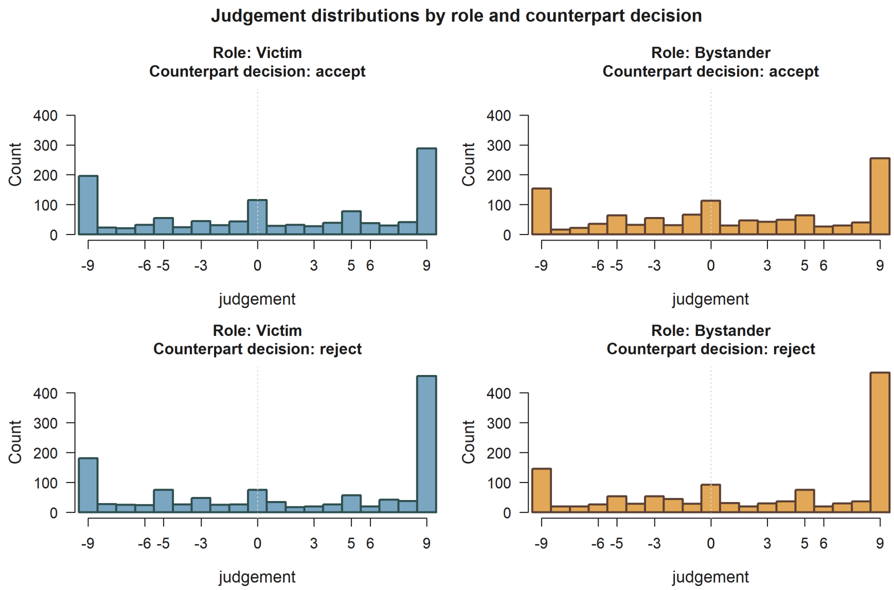

Figure 6: Observed judgement distributions by role and counterpart decision.

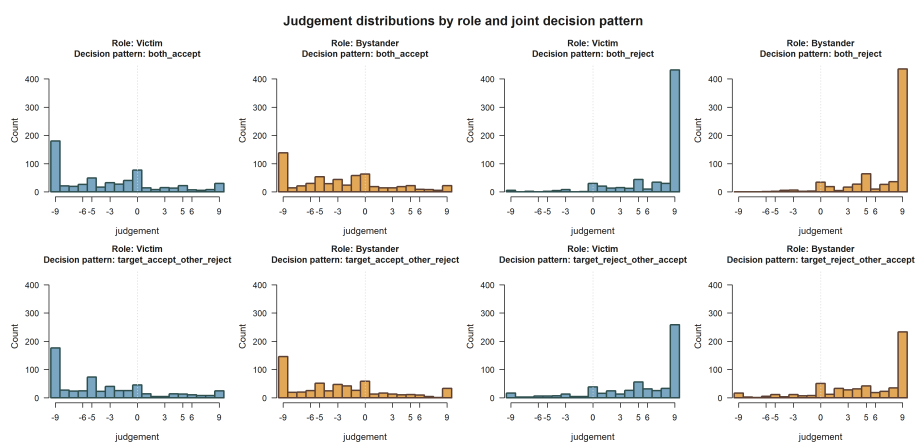

Figure 7: Observed judgement distributions by role and joint decision pattern.

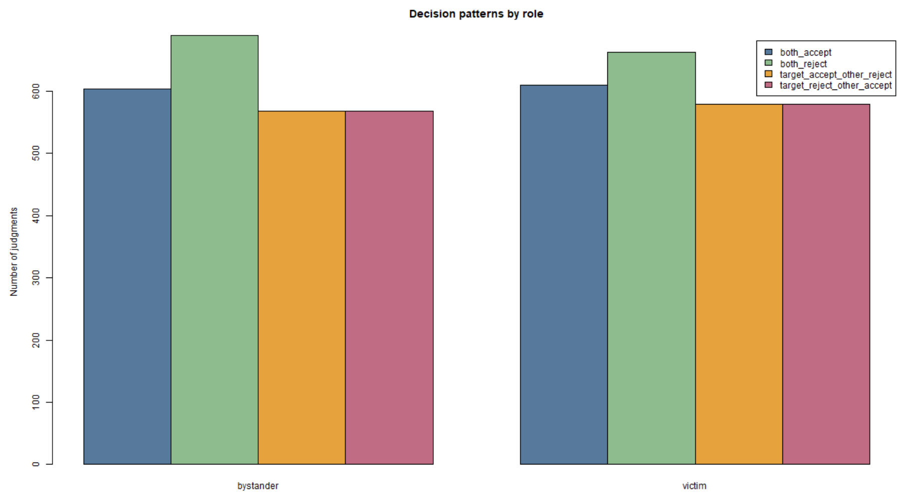

Figure 8: Observed decision-pattern counts by role.

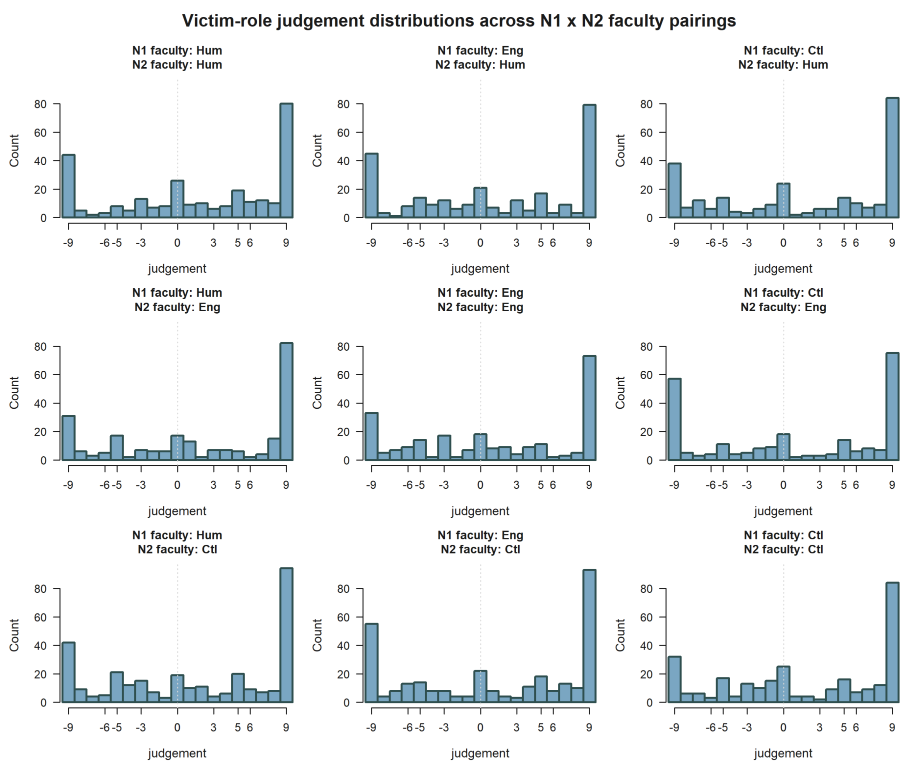

Figure 9: Victim-role judgement distributions across N1 x N2 faculty pairings.

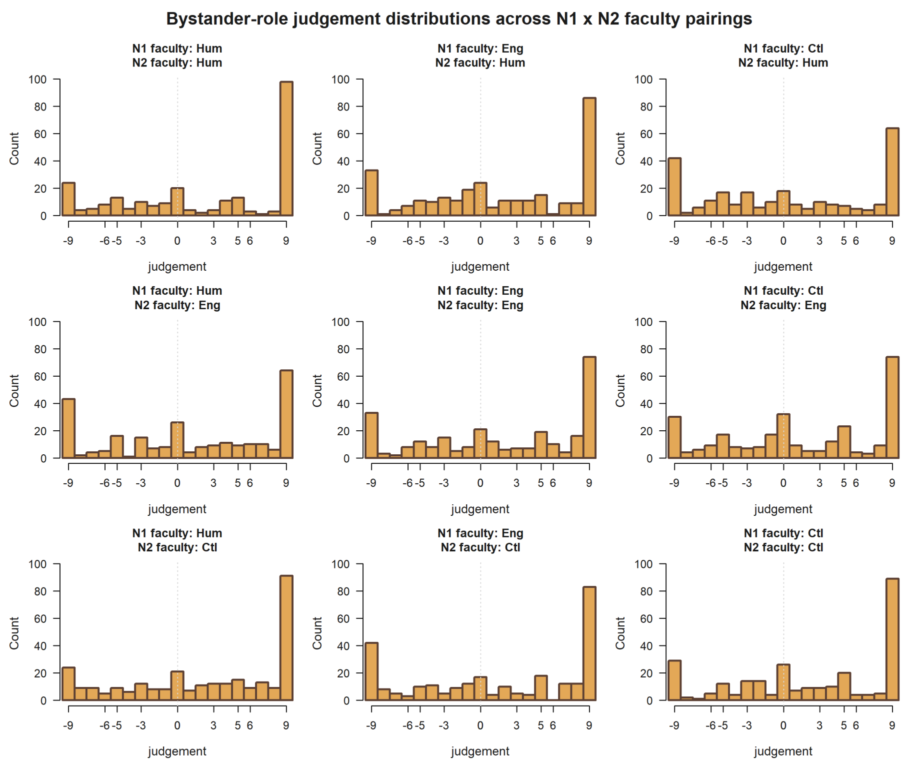

Figure 10: Bystander-role judgement distributions across N1 x N2 faculty pairings.

This figure shows how the raw judgement distribution varies across the full relational space defined by the faculty pairing of N1 and N2.

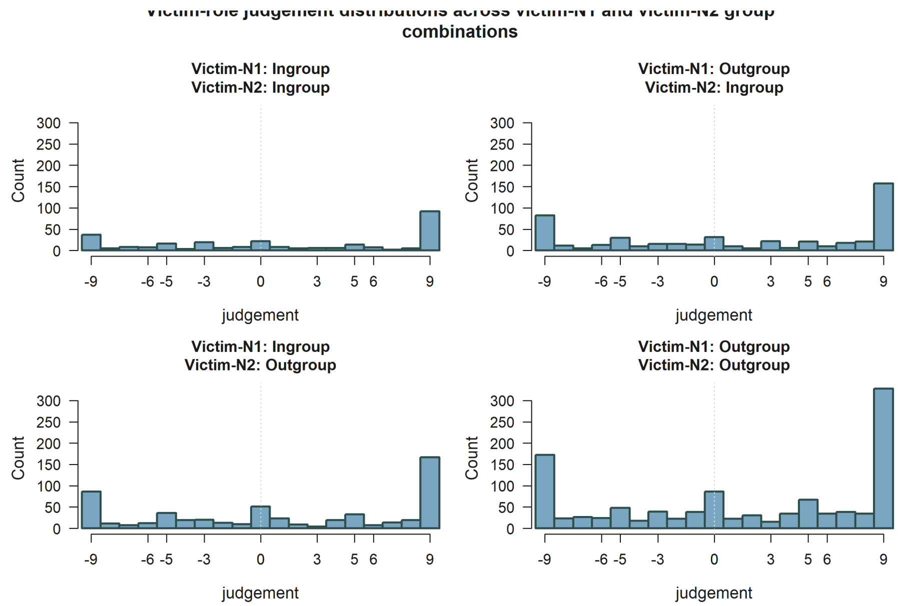

Figure 11: Victim-role judgement distributions across victim-N1 and victim-N2 ingroup/outgroup combinations.

This optional figure highlights how raw victim-role judgement varies across victim-centered ingroup/outgroup combinations.

This optional figure emphasizes bystander-side relational combinations directly tied to N1 and N2 group alignment.

This optional figure shows how bystander judgement co-varies with victim-centered relational combinations in the same observed rows.

11 Estimator fit summary

Bystander models use 2,420 observations from 243 participants. Victim models use 2,420 observations from 243 participants.

All production models use the estimator configuration shown below.

Table 11. Estimator configuration

Estimator	Session handling	Dependence
Two-sided Tobit	factor_session_fixed_effect	cluster_robust_id

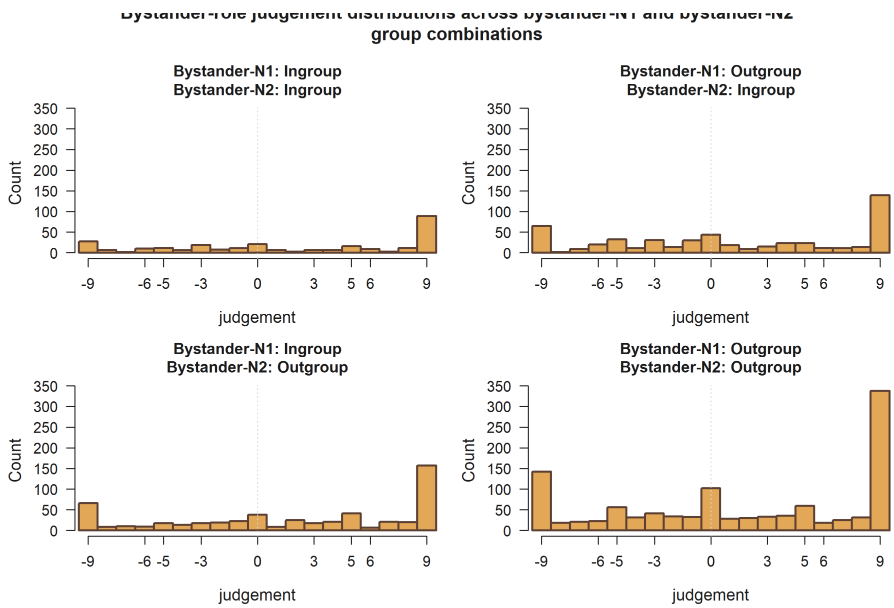

Figure 12: Bystander-role judgement distributions across bystander-N1 and bystander-N2 ingroup/outgroup combinations.

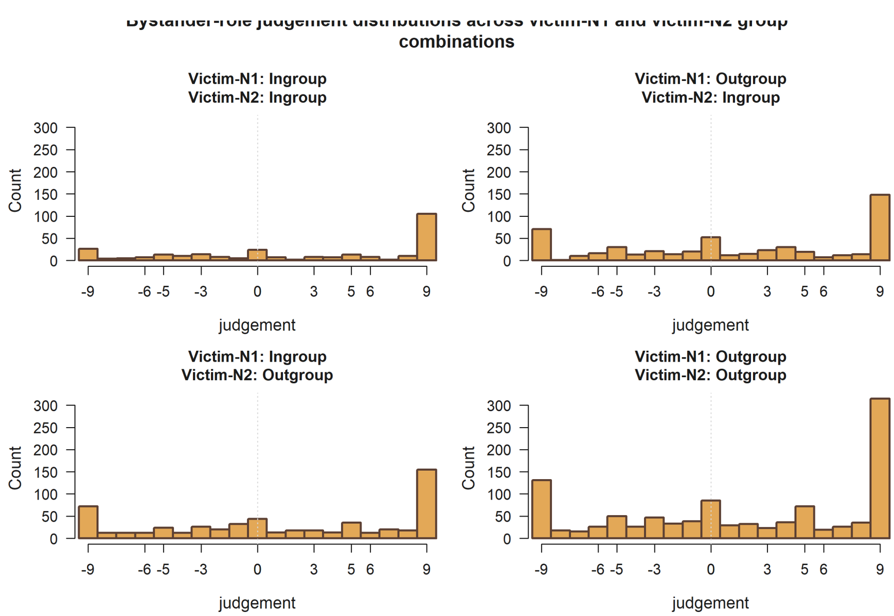

Figure 13: Bystander-role judgement distributions across victim-N1 and victim-N2 ingroup/outgroup combinations.

11.1 Model-level fit and censoring summary

11.1.1 Bystander models

Table 12. Bystander model fit and censoring summary

H	L. cens.	U. cens.	AIC	BIC	Sigma
H1	300	723	12558.800	12703.600	10.264
H2	300	723	12557.400	12725.400	10.242
H3	300	723	12569.200	12829.800	10.193
H4	300	723	11060.300	11199.300	6.917
H5	300	723	11080.200	11358.200	6.876

11.1.2 Victim models

Table 13. Victim model fit and censoring summary

H	L. cens.	U. cens.	AIC	BIC	Sigma
H1	377	744	12280.600	12425.300	11.564
H2	377	744	12297.600	12442.400	11.607
H3	377	744	12296.000	12510.300	11.541
H4	377	744	10747.900	10886.900	7.626
H5	377	744	10749.300	10980.900	7.570

12 Hypothesis significance summary by role

12.1 Victim

Table 14. Victim focal support terms (p < 0.10)

hypothesis	role	support
H1	Victim	Empathy: Perspective taking**
H2	Victim	None below p < 0.10
H3	Victim	Empathy: Fantasy*; Empathy: Perspective taking+
H4	Victim	Target accepted***; Other negotiator accepted***; Target accepted x Other accepted***

hypothesis	role	support
H5	Victim	Empathy: Fantasy*; Victim-N2 outgroup vs ingroup+; Target accepted*; **Other negotiator accepted***; Empathy: Fantasy x Victim-N1 outgroup vs ingroup+; Target accepted x Other accepted***

12.2 Bystander

Table 15. Bystander focal support terms (p < 0.10)

hypothesis	role	support
H1	Bystander	None below $p < 0.10$
H2	Bystander	Bystander-N2 outgroup vs ingroup+; Victim-N1 outgroup vs ingroup+; Victim-N2 outgroup vs ingroup+; Bystander-N1 outgroup vs ingroup x Bystander-N2 outgroup vs ingroup+; Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup+

hypothesis	role	support
H3	Bystander	Empathy: Empathic concern+; Empathy: Personal distress+; Victim-N1 outgroup vs ingroup+; Victim-N2 outgroup vs ingroup+; Bystander-N1 outgroup vs ingroup x Bystander-N2 outgroup vs ingroup+; Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup+; Empathy: Fantasy x Bystander-victim outgroup vs ingroup+; Empathy: Personal distress x Bystander-victim outgroup vs ingroup*
H4	Bystander	Target accepted***; Other negotiator accepted***; Target accepted x Other accepted***
H5	Bystander	Victim-N1 outgroup vs ingroup*; Victim-N2 outgroup vs ingroup*; Target accepted***; Other negotiator accepted***; Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup*; Empathy: Fantasy x Bystander-victim outgroup vs ingroup*; Target accepted x Other accepted***

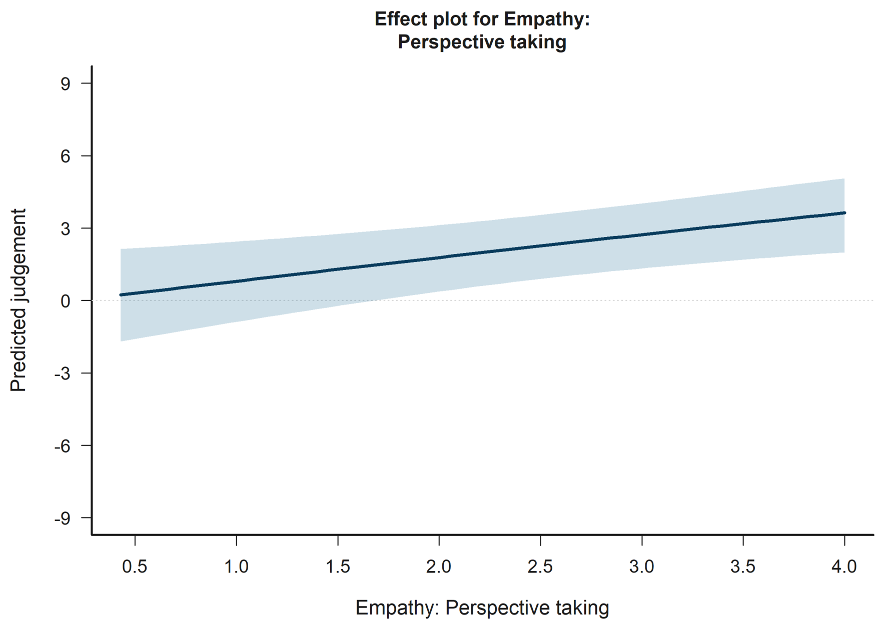

Figure 14: H1 Victim: model-implied predictions for Empathy: Perspective taking.

13 Significance-driven figures

13.1 H1 Victim: Empathy: Perspective taking

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

13.2 H2 Bystander: Bystander-N2 outgroup vs ingroup

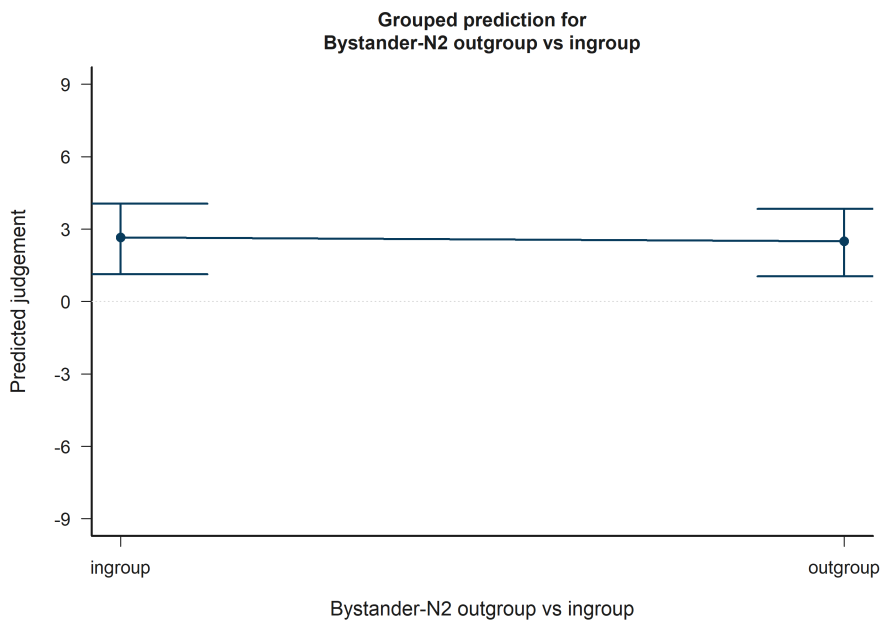

Figure 15: H2 Bystander: model-implied predictions for Bystander-N2 outgroup vs ingroup.

Across the displayed contrast, the model implies lower predicted judgement toward the right-hand side of the plot.

13.3 H2 Bystander: Victim-N1 outgroup vs ingroup

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

13.4 H2 Bystander: Victim-N2 outgroup vs ingroup

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

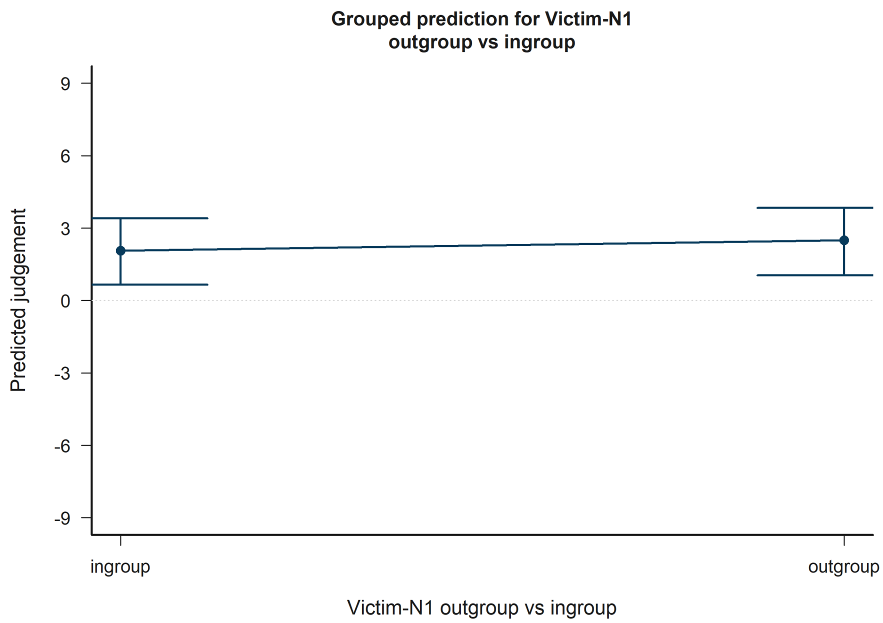

Figure 16: H2 Bystander: model-implied predictions for Victim-N1 outgroup vs ingroup.

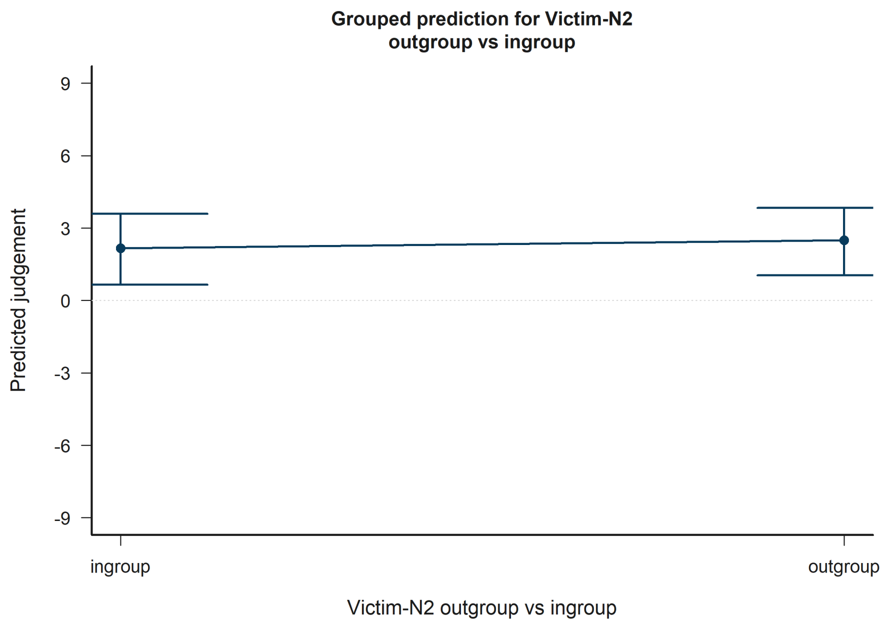

Figure 17: H2 Bystander: model-implied predictions for Victim-N2 outgroup vs ingroup.

13.5 H2 Bystander: Bystander-N1 outgroup vs ingroup x Bystander-N2 outgroup vs ingroup

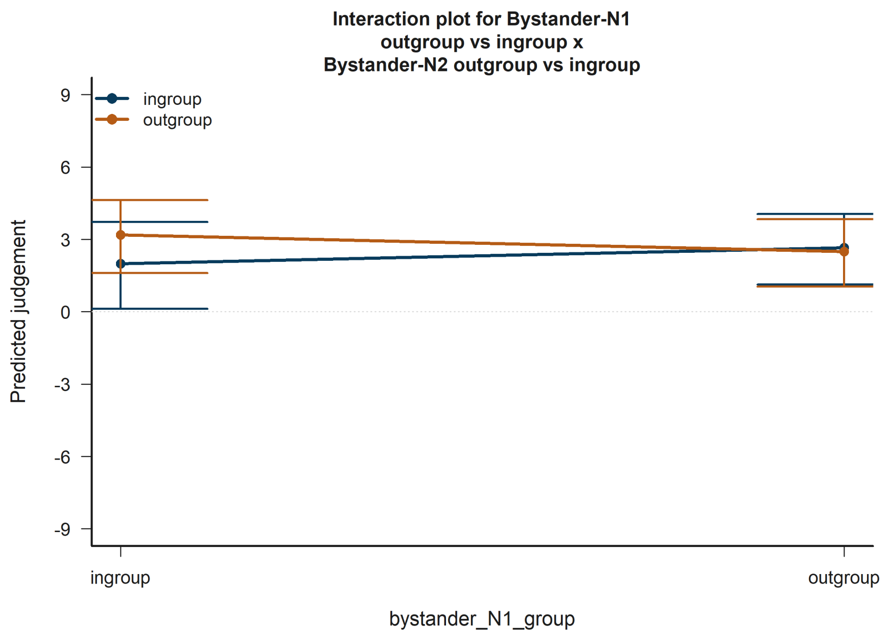

Figure 18: H2 Bystander: model-implied predictions for Bystander-N1 outgroup vs ingroup x Bystander-N2 outgroup vs ingroup.

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

13.6 H2 Bystander: Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

13.7 H3 Victim: Empathy: Fantasy

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

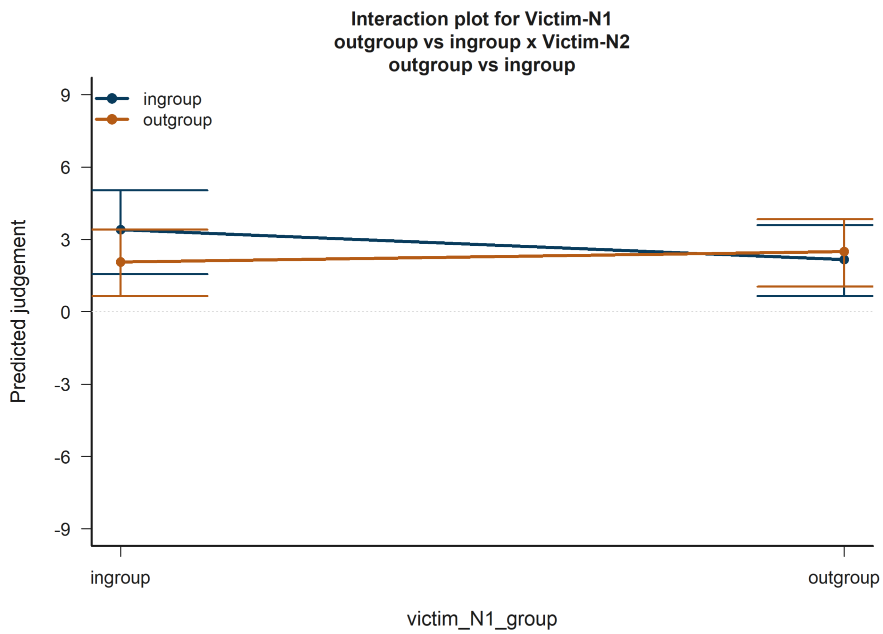

Figure 19: H2 Bystander: model-implied predictions for Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup.

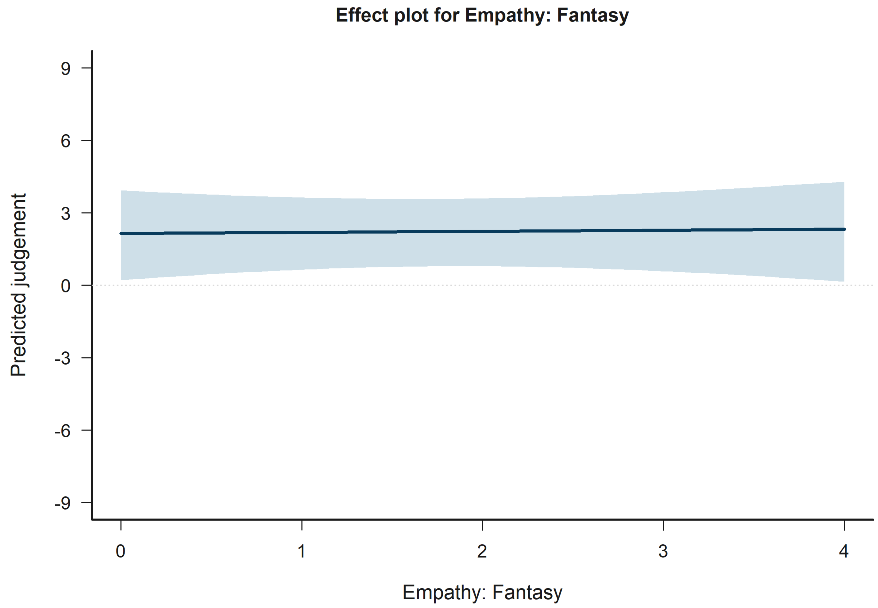

Figure 20: H3 Victim: model-implied predictions for Empathy: Fantasy.

13.8 H3 Victim: Empathy: Perspective taking

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

13.9 H3 Bystander: Empathy: Empathic concern

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

13.10 H3 Bystander: Empathy: Personal distress

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

13.11 H3 Bystander: Victim-N1 outgroup vs ingroup

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

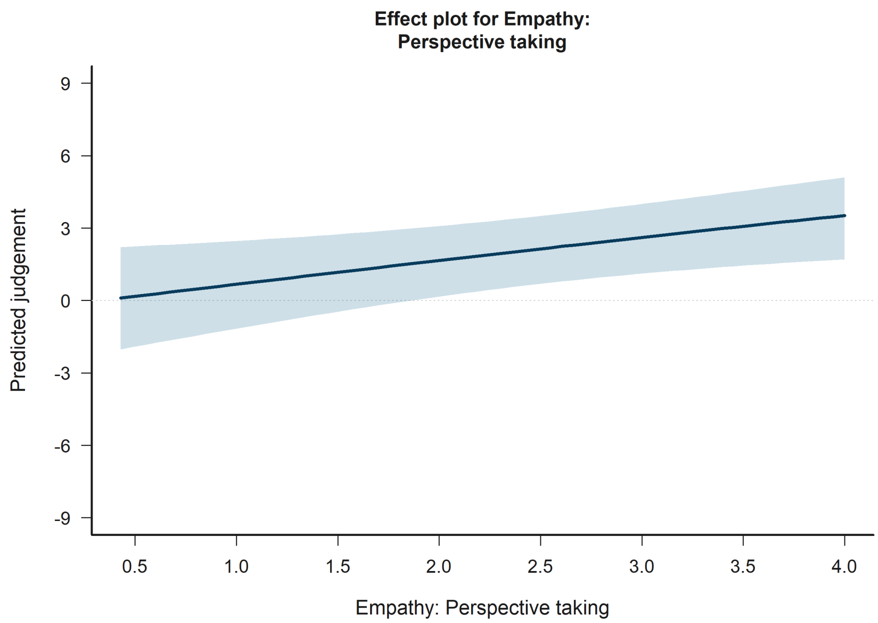

Figure 21: H3 Victim: model-implied predictions for Empathy: Perspective taking.

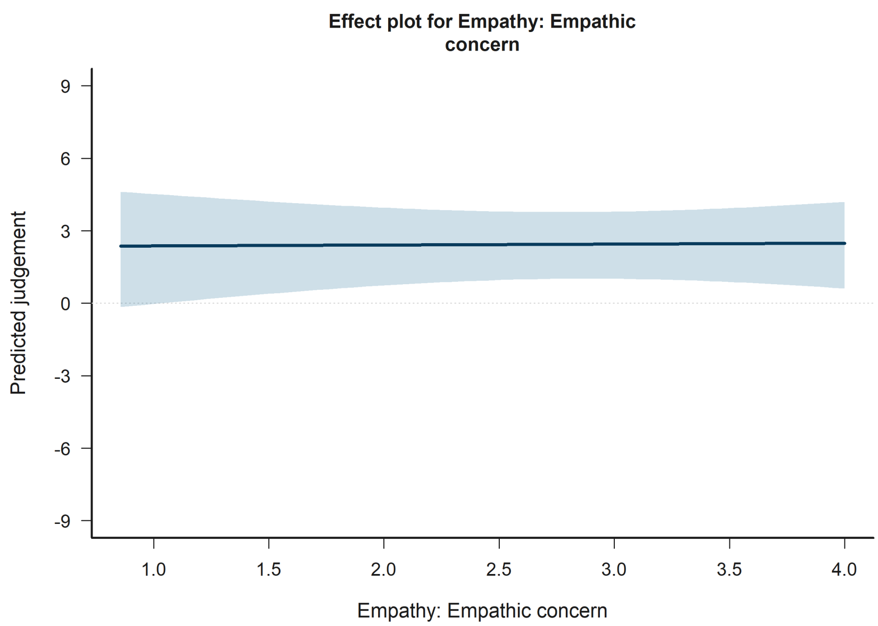

Figure 22: H3 Bystander: model-implied predictions for Empathy: Empathic concern.

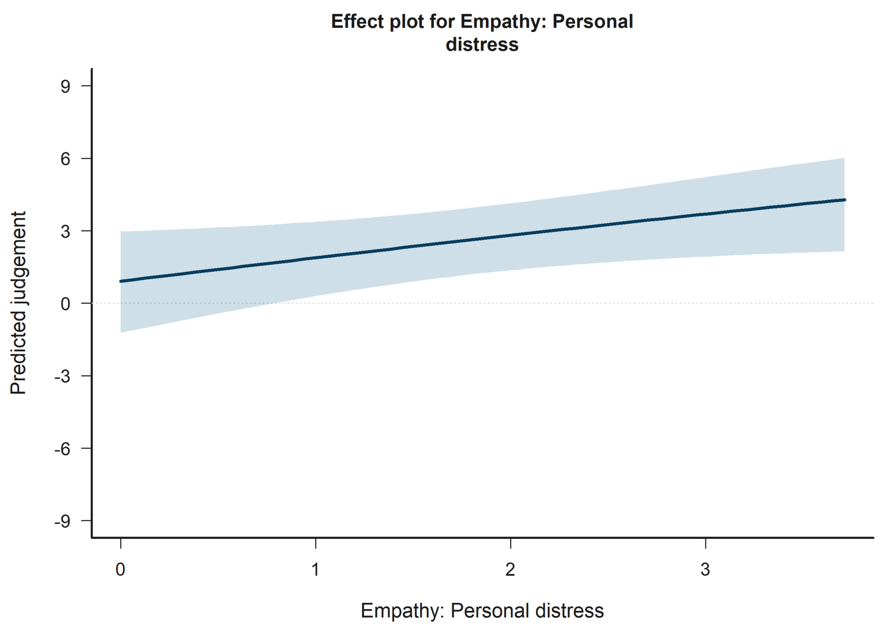

Figure 23: H3 Bystander: model-implied predictions for Empathy: Personal distress.

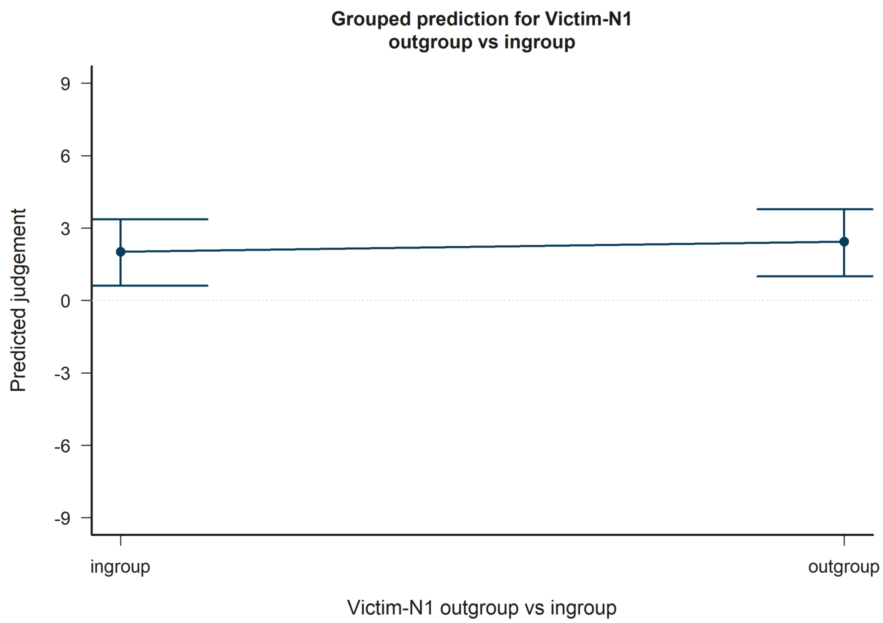

Figure 24: H3 Bystander: model-implied predictions for Victim-N1 outgroup vs ingroup.

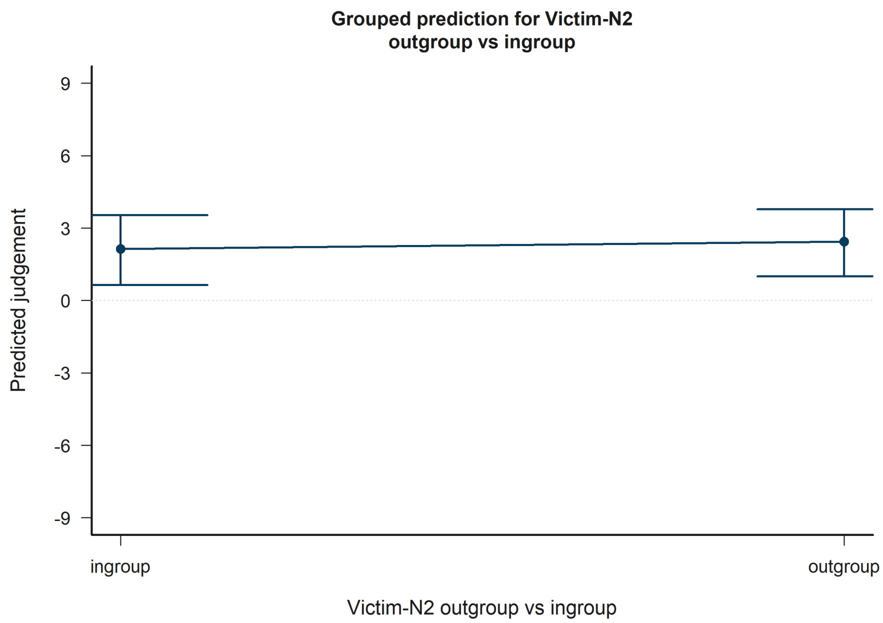

Figure 25: H3 Bystander: model-implied predictions for Victim-N2 outgroup vs ingroup.

13.12 H3 Bystander: Victim-N2 outgroup vs ingroup

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

13.13 H3 Bystander: Bystander-N1 outgroup vs ingroup x Bystander-N2 outgroup vs ingroup

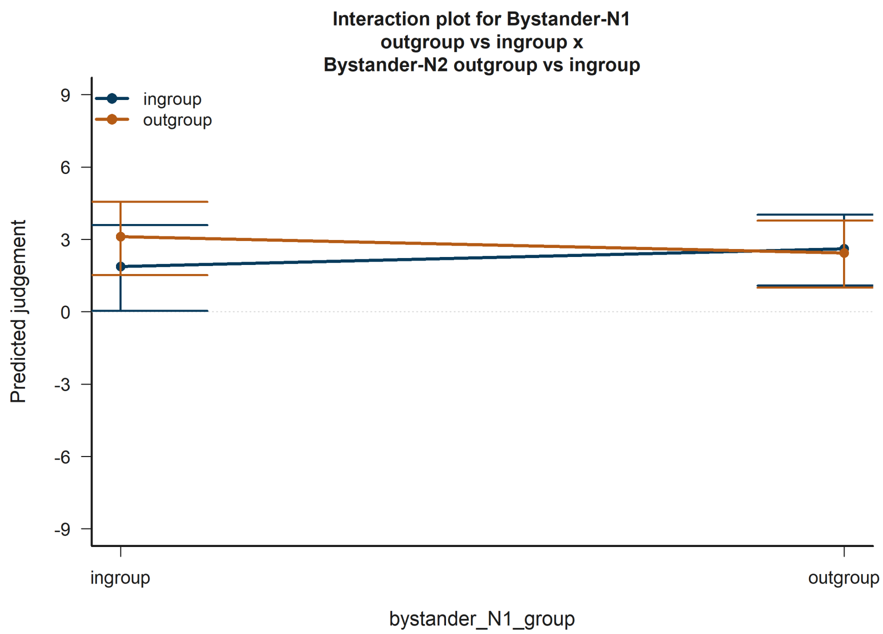

Figure 26: H3 Bystander: model-implied predictions for Bystander-N1 outgroup vs ingroup x Bystander-N2 outgroup vs ingroup.

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

13.14 H3 Bystander: Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

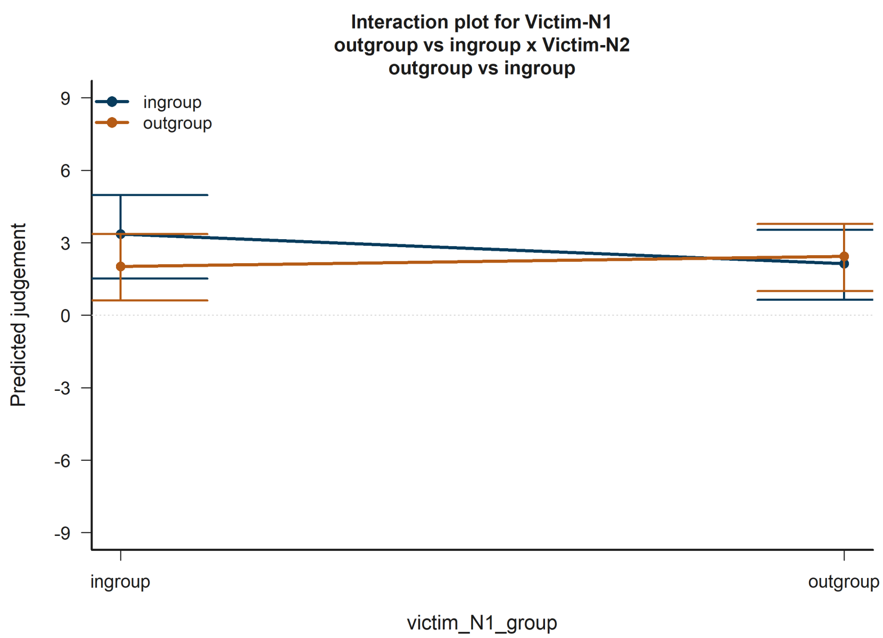

Figure 27: H3 Bystander: model-implied predictions for Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup.

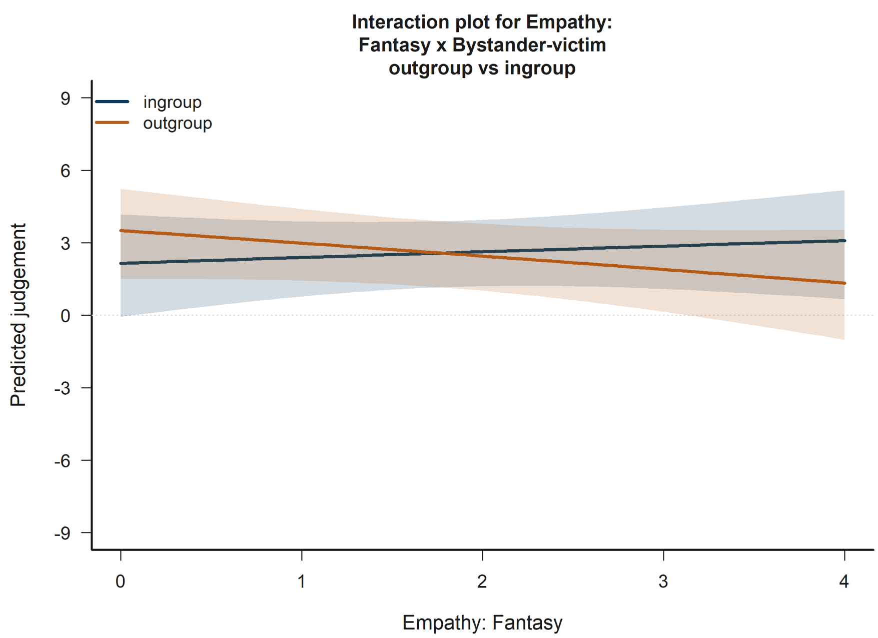

Figure 28: H3 Bystander: model-implied predictions for Empathy: Fantasy x Bystander-victim outgroup vs ingroup.

13.15 H3 Bystander: Empathy: Fantasy x Bystander-victim outgroup vs ingroup

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

13.16 H3 Bystander: Empathy: Personal distress x Bystander-victim outgroup vs ingroup

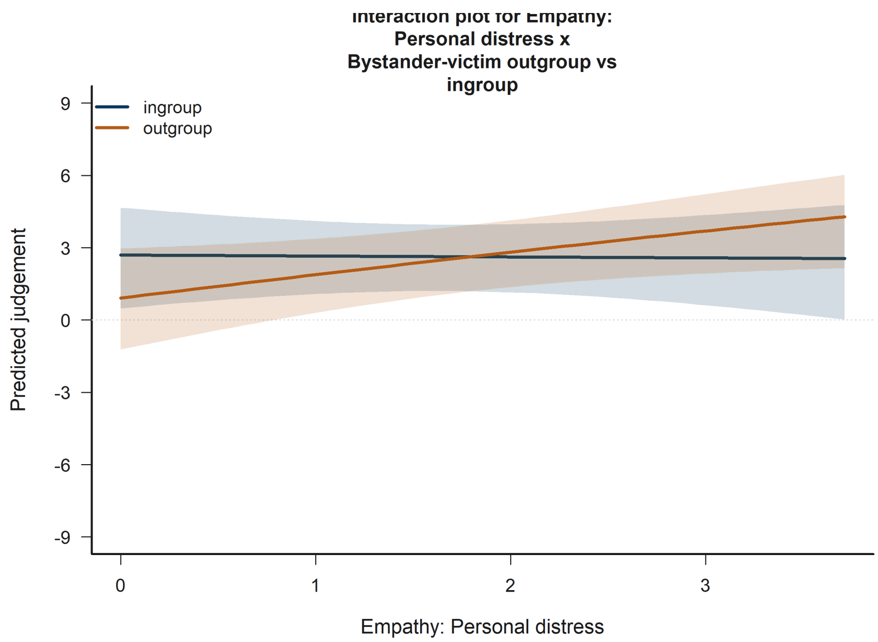

Figure 29: H3 Bystander: model-implied predictions for Empathy: Personal distress x Bystander-victim outgroup vs ingroup.

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

13.17 H4 Victim: Target accepted

Across the displayed contrast, the model implies lower predicted judgement toward the right-hand side of the plot.

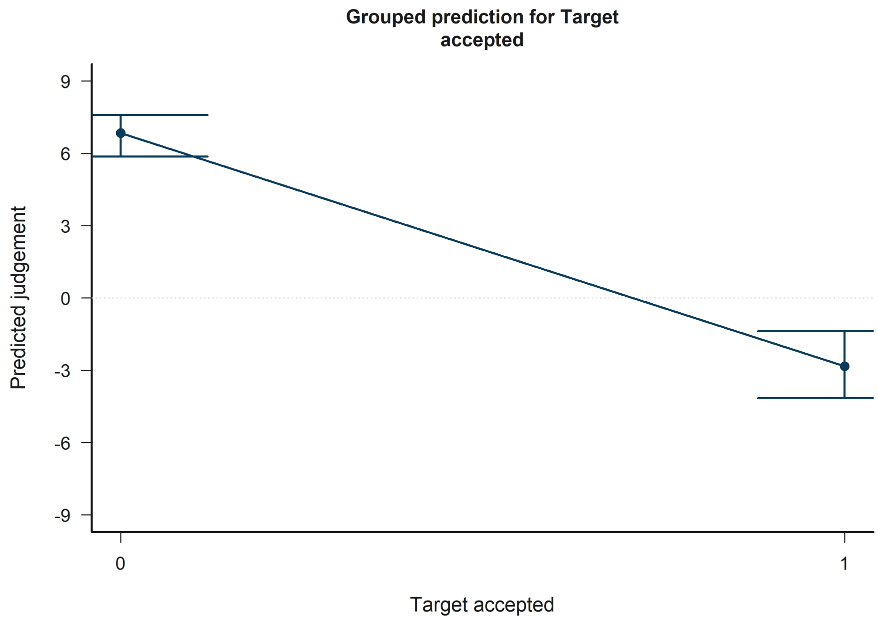

Figure 30: H4 Victim: model-implied predictions for Target accepted.

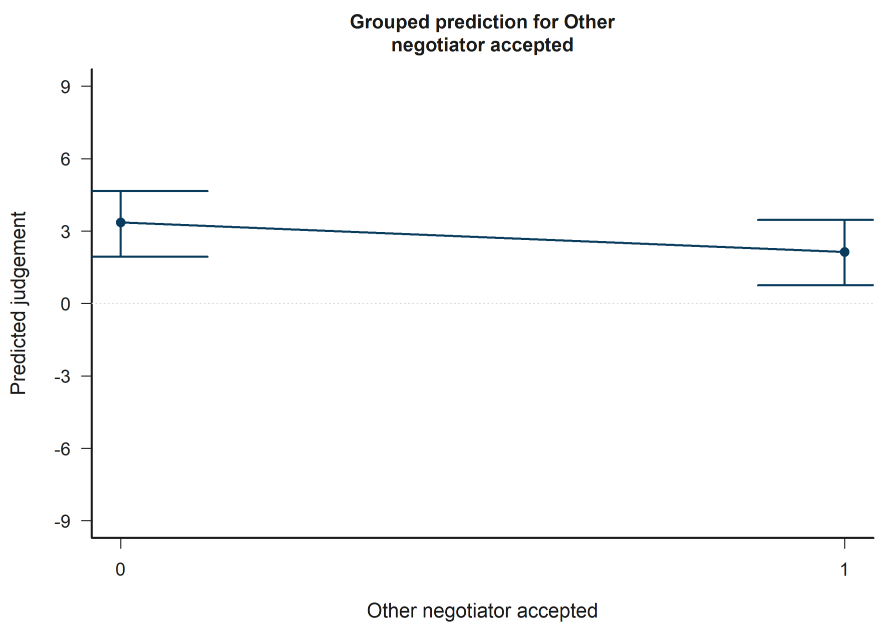

Figure 31: H4 Victim: model-implied predictions for Other negotiator accepted.

13.18 H4 Victim: Other negotiator accepted

Across the displayed contrast, the model implies lower predicted judgement toward the right-hand side of the plot.

13.19 H4 Victim: Target accepted x Other accepted

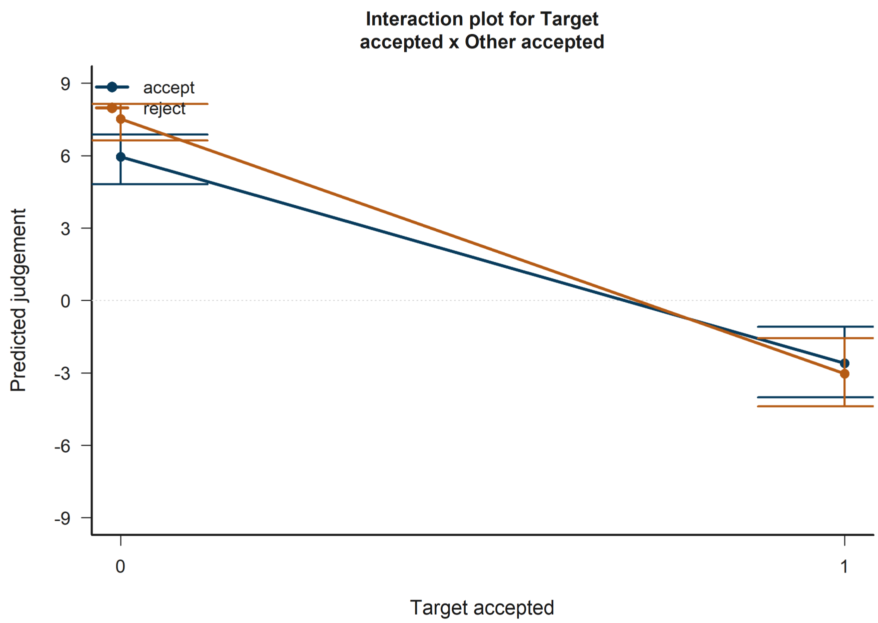

Figure 32: H4 Victim: model-implied predictions for Target accepted x Other accepted.

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

13.20 H4 Bystander: Target accepted

Across the displayed contrast, the model implies lower predicted judgement toward the right-hand side of the plot.

13.21 H4 Bystander: Other negotiator accepted

Across the displayed contrast, the model implies lower predicted judgement toward the right-hand side of the plot.

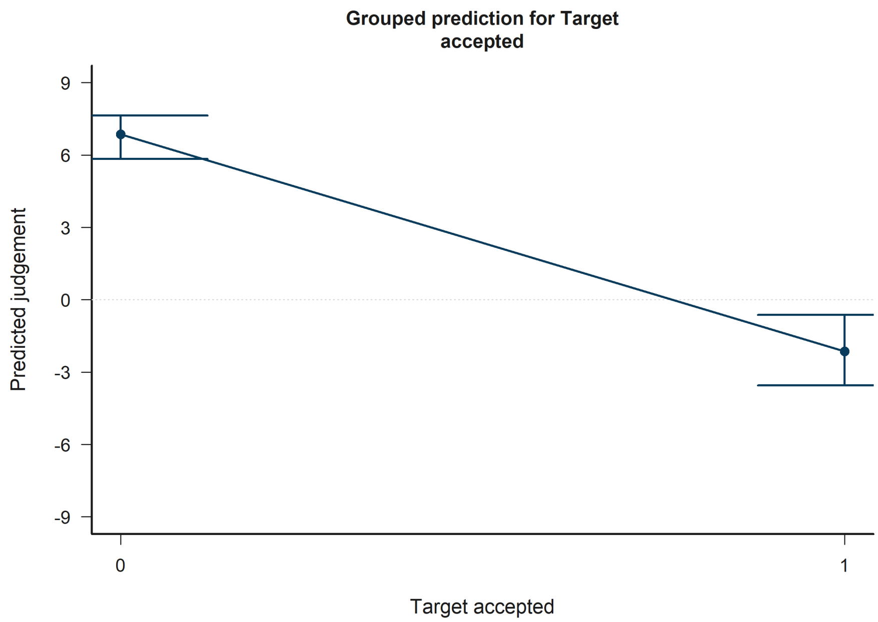

Figure 33: H4 Bystander: model-implied predictions for Target accepted.

13.22 H4 Bystander: Target accepted x Other accepted

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

13.23 H5 Victim: Empathy: Fantasy

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

13.24 H5 Victim: Victim-N2 outgroup vs ingroup

Across the displayed contrast, the model implies lower predicted judgement toward the right-hand side of the plot.

13.25 H5 Victim: Target accepted

Across the displayed contrast, the model implies lower predicted judgement toward the right-hand side of the plot.

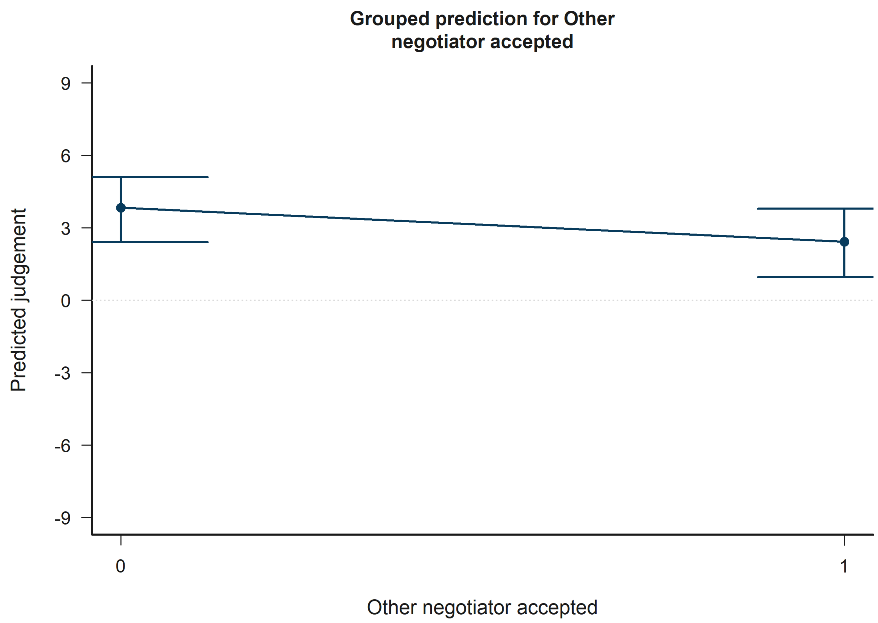

Figure 34: H4 Bystander: model-implied predictions for Other negotiator accepted.

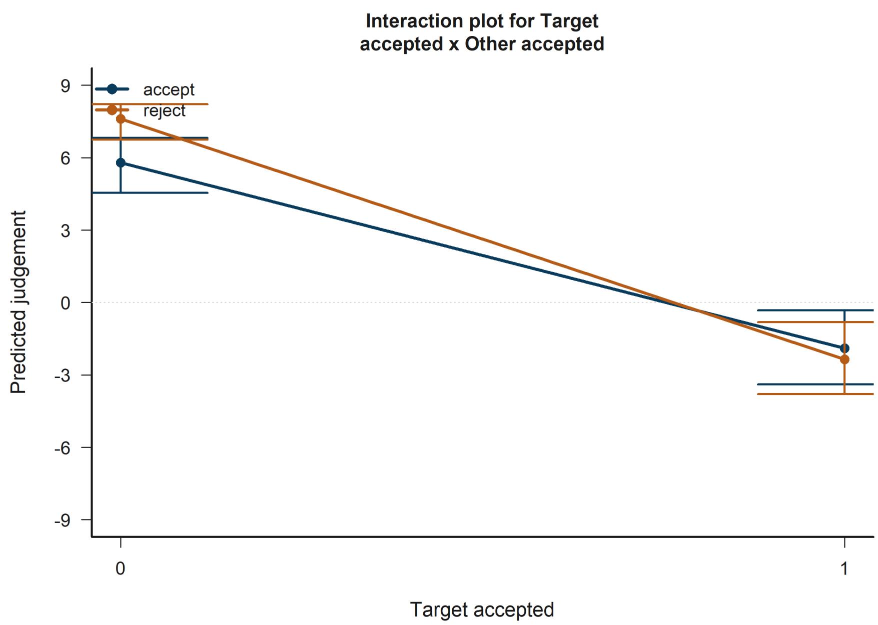

Figure 35: H4 Bystander: model-implied predictions for Target accepted x Other accepted.

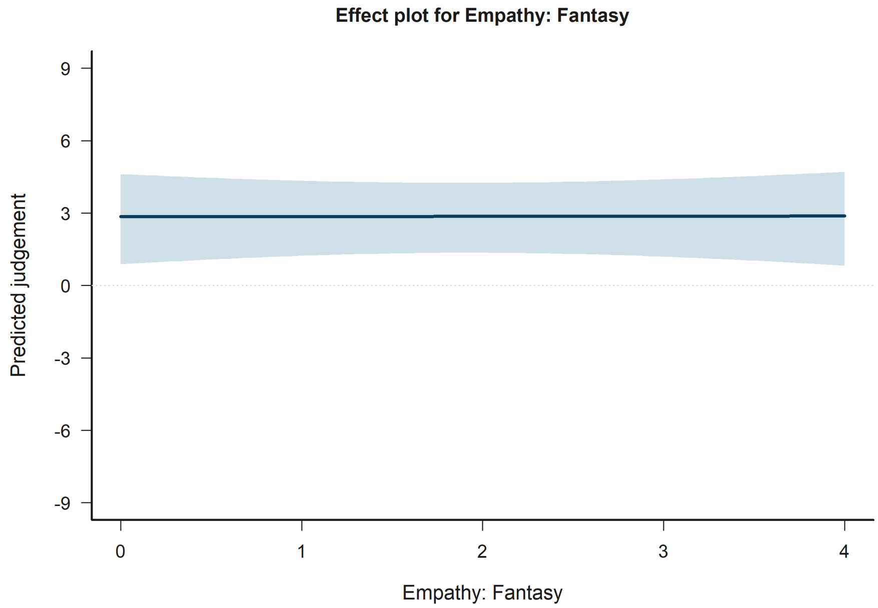

Figure 36: H5 Victim: model-implied predictions for Empathy: Fantasy.

13.26 H5 Victim: Other negotiator accepted

Across the displayed contrast, the model implies lower predicted judgement toward the right-hand side of the plot.

13.27 H5 Victim: Empathy: Fantasy x Victim-N1 outgroup vs ingroup

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

13.28 H5 Victim: Target accepted x Other accepted

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

13.29 H5 Bystander: Victim-N1 outgroup vs ingroup

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

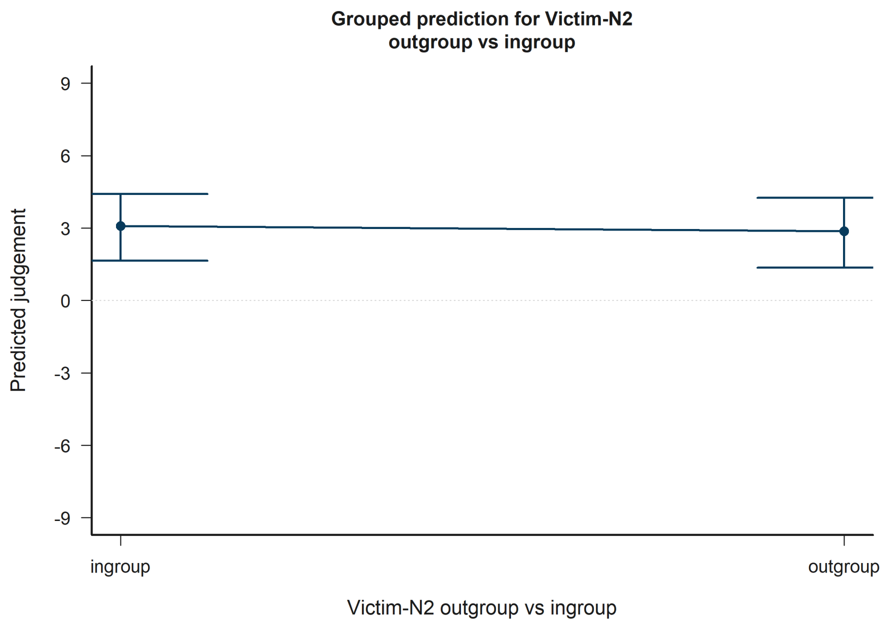

Figure 37: H5 Victim: model-implied predictions for Victim-N2 outgroup vs ingroup.

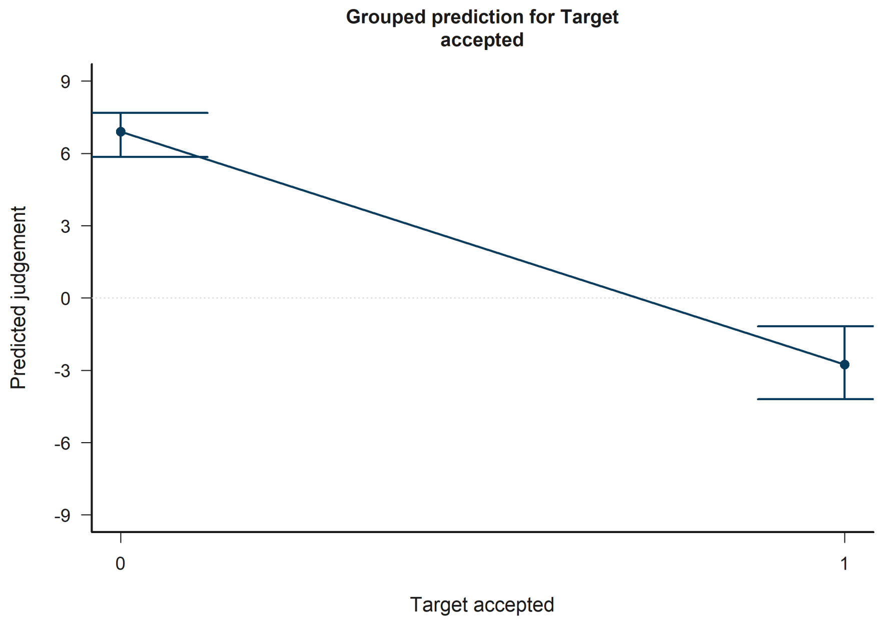

Figure 38: H5 Victim: model-implied predictions for Target accepted.

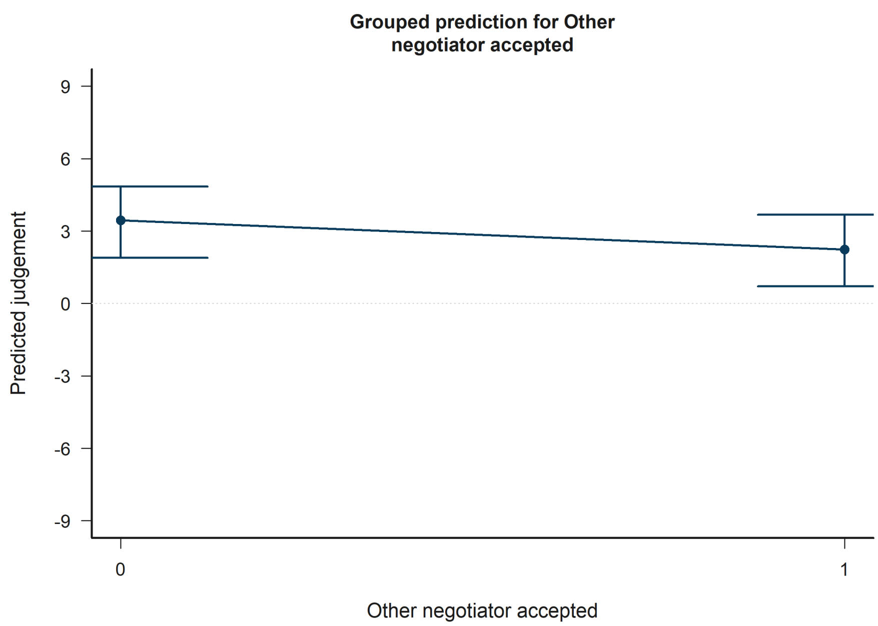

Figure 39: H5 Victim: model-implied predictions for Other negotiator accepted.

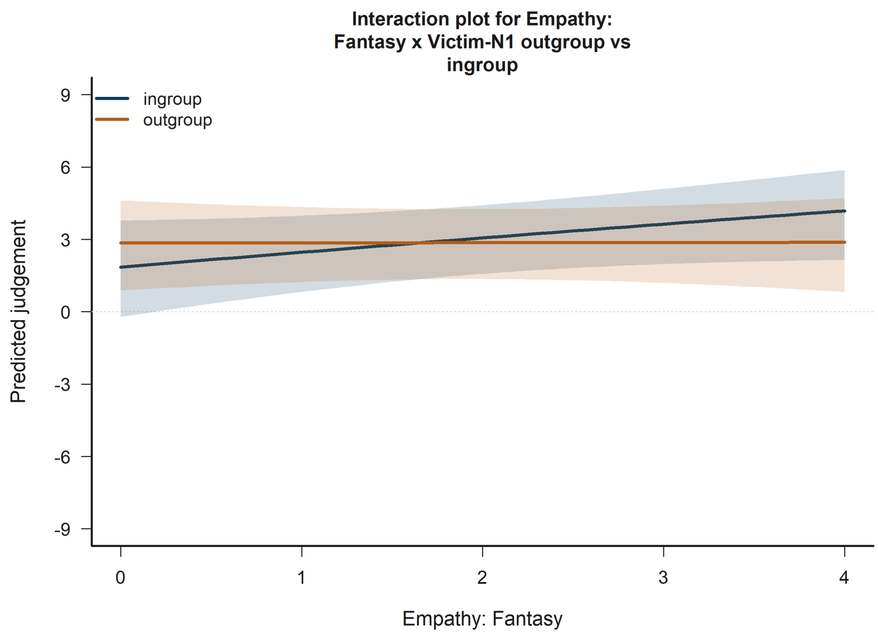

Figure 40: H5 Victim: model-implied predictions for Empathy: Fantasy x Victim-N1 outgroup vs ingroup.

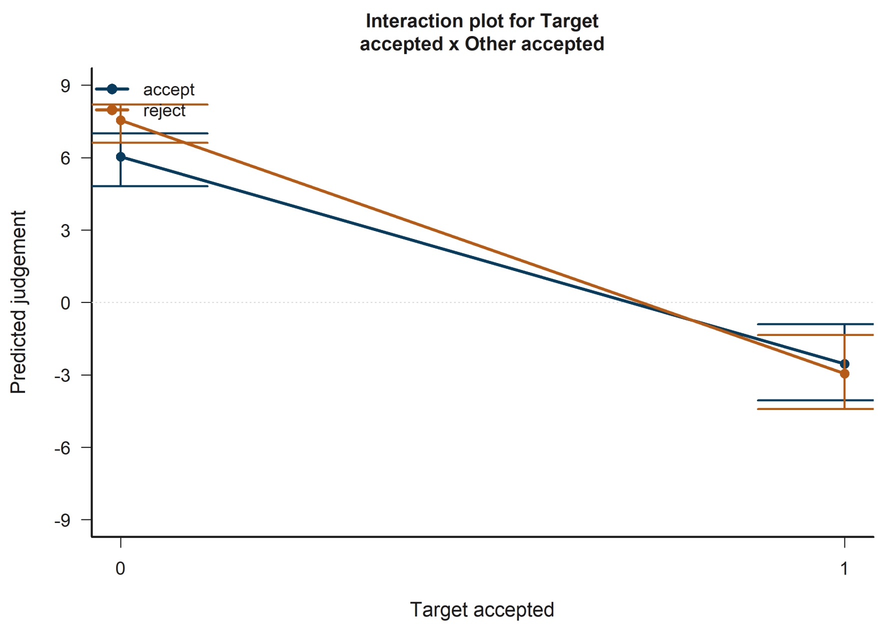

Figure 41: H5 Victim: model-implied predictions for Target accepted x Other accepted.

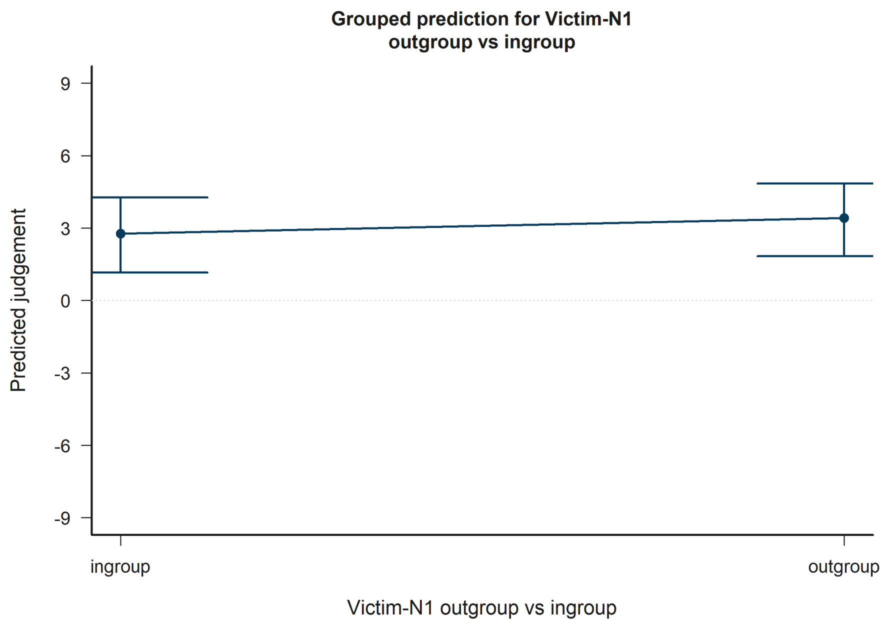

Figure 42: H5 Bystander: model-implied predictions for Victim-N1 outgroup vs ingroup.

13.30 H5 Bystander: Victim-N2 outgroup vs ingroup

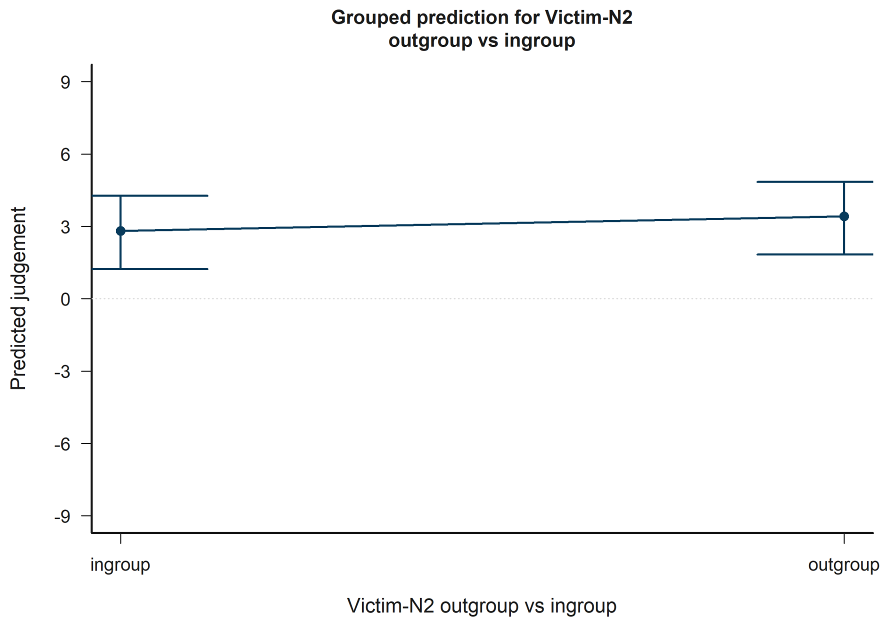

Figure 43: H5 Bystander: model-implied predictions for Victim-N2 outgroup vs ingroup.

Across the displayed contrast, the model implies higher predicted judgement toward the right-hand side of the plot.

13.31 H5 Bystander: Target accepted

Across the displayed contrast, the model implies lower predicted judgement toward the right-hand side of the plot.

13.32 H5 Bystander: Other negotiator accepted

Across the displayed contrast, the model implies lower predicted judgement toward the right-hand side of the plot.

13.33 H5 Bystander: Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

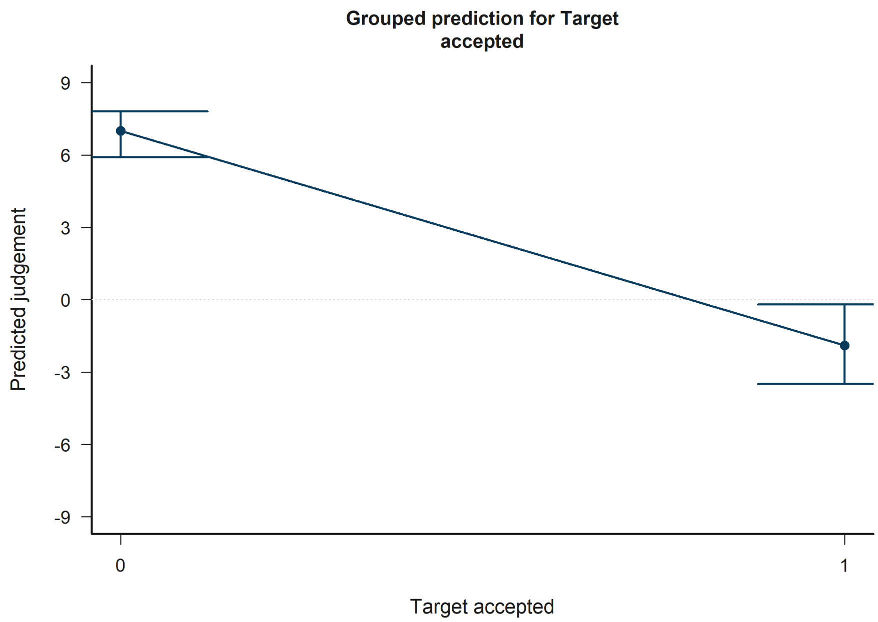

Figure 44: H5 Bystander: model-implied predictions for Target accepted.

13.34 H5 Bystander: Empathy: Fantasy x Bystander-victim outgroup vs ingroup

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

13.35 H5 Bystander: Target accepted x Other accepted

The plotted fitted lines and shaded 95% confidence bands summarize how predicted judgement changes across the focal term while holding remaining covariates at their reference profile.

14 Full coefficient tables and interpretation summary

14.1 H1 Victim coefficient table

Table 16. H1 Victim coefficient estimates

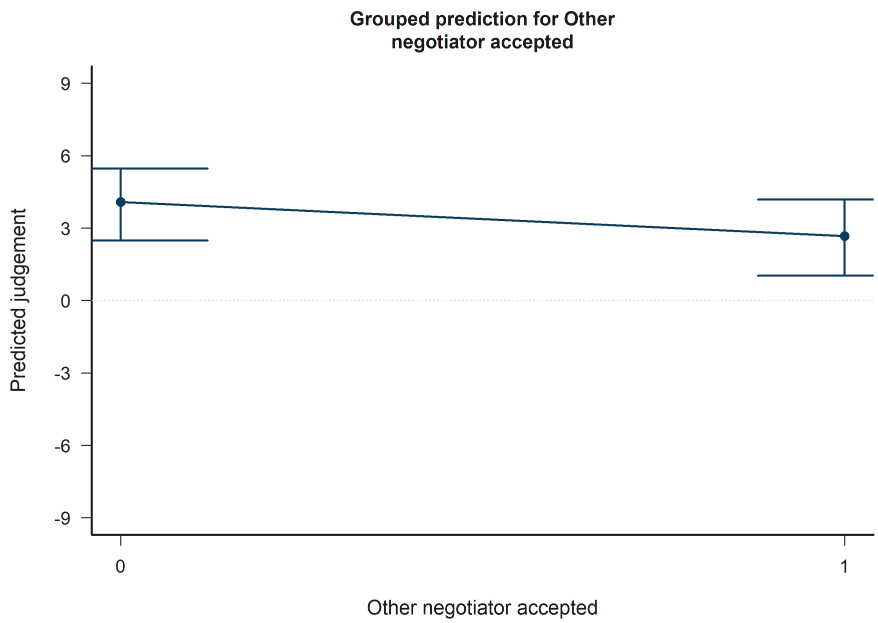

Figure 45: H5 Bystander: model-implied predictions for Other negotiator accepted.

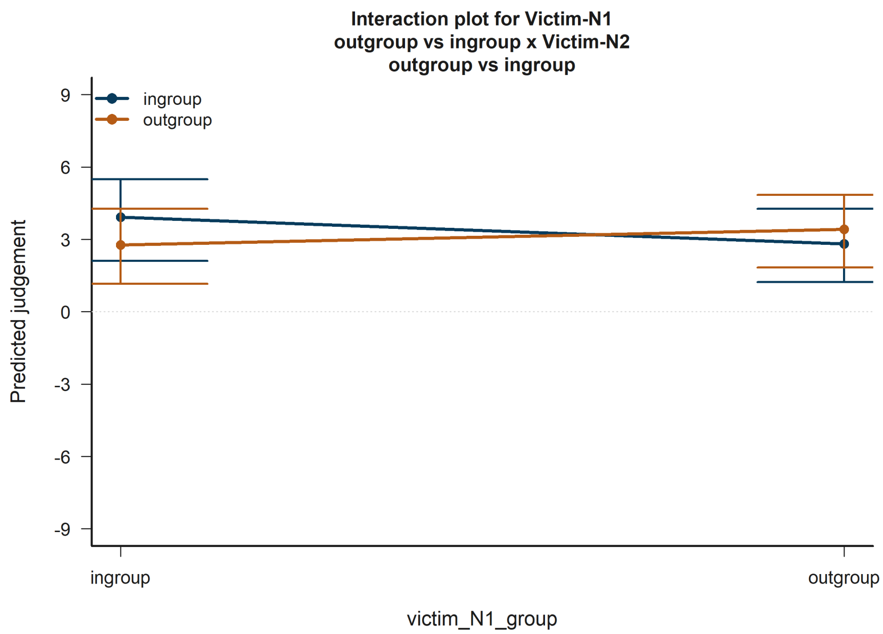

Figure 46: H5 Bystander: model-implied predictions for Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup.

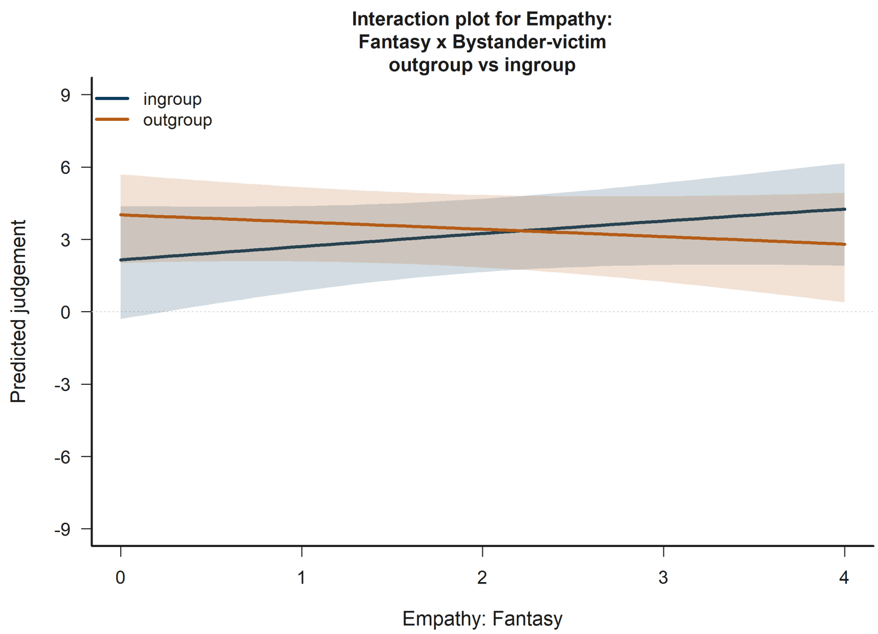

Figure 47: H5 Bystander: model-implied predictions for Empathy: Fantasy x Bystander-victim outgroup vs ingroup.

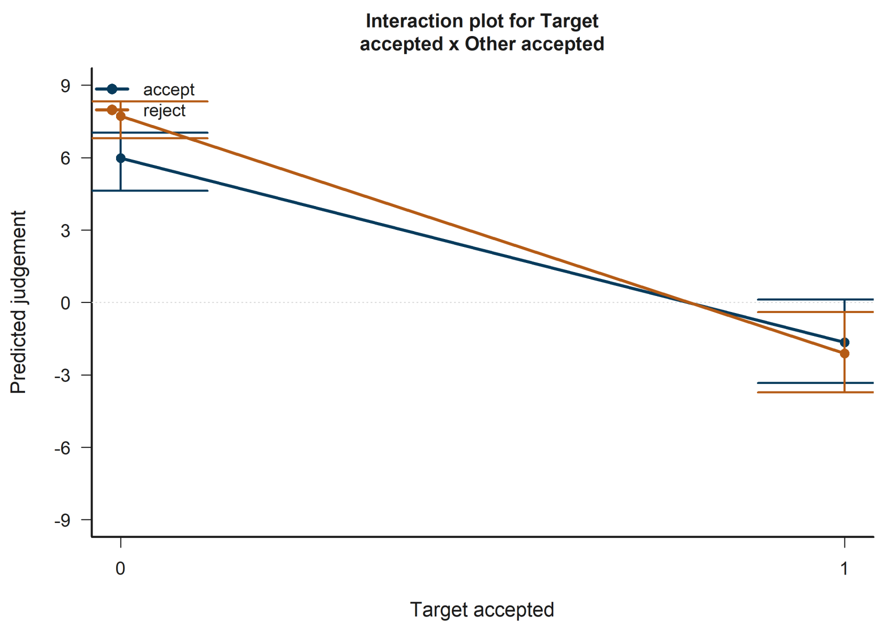

Figure 48: H5 Bystander: model-implied predictions for Target accepted x Other accepted.

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Intercept	-0.874	3.297	-7.337	5.589	0.791
Empathy: Fantasy	0.644	0.524	-0.383	1.672	0.219
Empathy: Empathic concern	-0.850	0.751	-2.322	0.623	0.258
Empathy: Perspective taking	1.773	0.614	0.570	2.975	0.004**
Empathy: Personal distress	0.035	0.588	-1.117	1.188	0.952
Age	-0.025	0.133	-0.285	0.235	0.850
Socioeconomic status	0.451	0.344	-0.224	1.126	0.190
Woman participant	0.211	0.812	-1.380	1.803	0.795
Participant faculty: Engineering vs Humanities	1.352	0.803	-0.222	2.925	0.092+
Tobit log-scale	2.448	0.057	2.336	2.560	<0.001***

The H1 Victim model shows focal evidence for one hypothesis terms. Empathy: Perspective taking is associated with higher predicted judgement (estimate = 1.77, p = 0.004**). Session dummies remain in the fitted estimator for adjustment but are intentionally omitted from the substantive narrative.

14.2 H1 Bystander coefficient table

Table 17. H1 Bystander coefficient estimates

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Intercept	-3.438	3.171	-9.654	2.777	0.278
Empathy: Fantasy	0.111	0.446	-0.763	0.984	0.804
Empathy: Empathic concern	-0.859	0.638	-2.110	0.391	0.178

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Empathy: Perspective taking	0.573	0.560	-0.524	1.671	0.306
Empathy: Personal distress	0.167	0.561	-0.934	1.267	0.766
Age	0.361	0.103	0.160	0.563	<0.001***
Socioeconomic status	0.491	0.299	-0.096	1.078	0.101
Woman participant	-0.810	0.704	-2.190	0.569	0.250
Participant faculty: Engineering vs Humanities	1.005	0.626	-0.222	2.232	0.109
Tobit log-scale	2.329	0.057	2.217	2.440	<0.001***

In the H1 Bystander model, no focal hypothesis term reached $p < 0.10$. The report therefore retains the coefficient table for auditability but does not attach a significance-driven substantive interpretation beyond the descriptive prediction plots.

H2 reminder: `N1/N2` are structural slots reconstructed within each row, not fixed aliases of `target/other`; `group_target` and `group_other` are legacy source-audit fields, while active H2 models use reconstructed relational predictors.

14.3 H2 Victim coefficient table

Table 18. H2 Victim coefficient estimates

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Intercept	2.846	3.504	-4.023	9.715	0.417
Victim-N1 outgroup vs ingroup	-0.232	1.364	-2.905	2.441	0.865
Victim-N2 outgroup vs ingroup	-0.424	1.310	-2.992	2.144	0.746

term_label	estimate	std_error	conf_low	conf_high	p_value_display
N1/N2 same faculty vs different	0.359	0.849	-1.305	2.023	0.672
Age	-0.020	0.133	-0.281	0.241	0.881
Socioeconomic status	0.525	0.358	-0.176	1.227	0.142
Woman participant	0.327	0.764	-1.171	1.824	0.669
Participant faculty: Engineering vs Humanities	1.526	0.817	-0.076	3.128	0.062+
Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup	0.133	1.792	-3.379	3.646	0.941
Tobit log-scale	2.452	0.057	2.340	2.563	<0.001***

In the H2 Victim model, no focal hypothesis term reached $p < 0.10$. The report therefore retains the coefficient table for auditability but does not attach a significance-driven substantive interpretation beyond the descriptive prediction plots.

14.4 H2 Bystander coefficient table

Table 19. H2 Bystander coefficient estimates

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Intercept	-2.381	3.001	-8.263	3.501	0.428
Bystander-victim outgroup vs ingroup	-0.315	0.495	-1.285	0.656	0.525

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Bystander-N1 outgroup vs ingroup	1.132	1.078	-0.980	3.245	0.294
Bystander-N2 outgroup vs ingroup	2.085	1.120	-0.110	4.280	0.063+
Victim-N1 outgroup vs ingroup	-2.168	1.227	-4.573	0.237	0.077+
Victim-N2 outgroup vs ingroup	-2.331	1.214	-4.710	0.049	0.055+
N1/N2 same faculty vs different	0.786	0.831	-0.844	2.416	0.344
Age	0.325	0.101	0.126	0.524	0.001**
Socioeconomic status	0.466	0.299	-0.120	1.052	0.119
Woman participant	-0.902	0.661	-2.198	0.394	0.173
Participant faculty: Engineering vs Humanities	1.064	0.646	-0.202	2.331	0.099+
Bystander-N1 outgroup vs ingroup x Bystander-N2 outgroup vs ingroup	-2.359	1.401	-5.104	0.386	0.092+

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup	2.884	1.578	-0.210	5.977	0.068+
Tobit log-scale	2.326	0.057	2.215	2.438	<0.001***

The H2 Bystander model shows focal evidence for 5 hypothesis terms. Victim-N2 outgroup vs ingroup is associated with lower predicted judgement (estimate = -2.33, p = 0.055+). Bystander-N2 outgroup vs ingroup is associated with higher predicted judgement (estimate = 2.08, p = 0.063+). Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup is associated with higher predicted judgement (estimate = 2.88, p = 0.068+). Session dummies remain in the fitted estimator for adjustment but are intentionally omitted from the substantive narrative.

14.5 H3 Victim coefficient table

Table 20. H3 Victim coefficient estimates

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Intercept	-4.295	4.224	-12.574	3.984	0.309
Empathy: Fantasy	2.044	0.859	0.360	3.727	0.017*
Empathy: Empathic concern	-0.137	1.104	-2.302	2.027	0.901
Empathy: Perspective taking	1.678	0.980	-0.242	3.598	0.087+
Empathy: Personal distress	-0.416	1.038	-2.451	1.619	0.689
Victim-N1 outgroup vs ingroup	1.415	2.894	-4.257	7.087	0.625
Victim-N2 outgroup vs ingroup	2.983	3.200	-3.288	9.255	0.351

term_label	estimate	std_error	conf_low	conf_high	p_value_display
N1/N2 same faculty vs different	0.297	0.845	-1.360	1.954	0.725
Age	-0.025	0.132	-0.284	0.235	0.853
Socioeconomic status	0.472	0.344	-0.203	1.146	0.171
Woman participant	0.248	0.817	-1.352	1.849	0.761
Participant faculty: Engineering vs Humanities	1.519	0.812	-0.073	3.110	0.061+
Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup	0.328	1.780	-3.161	3.816	0.854
Empathy: Fantasy x Victim-N1 outgroup vs ingroup	-0.829	0.752	-2.302	0.644	0.270
Empathy: Fantasy x Victim-N2 outgroup vs ingroup	-1.132	0.883	-2.863	0.599	0.200
Empathy: Empathic concern x Victim-N1 outgroup vs ingroup	0.380	1.113	-1.802	2.562	0.733

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Empathy: Empathic concern x Victim-N2 outgroup vs ingroup	-1.357	1.116	-3.544	0.831	0.224
Empathy: Perspective taking x Victim-N1 outgroup vs ingroup	-0.388	1.042	-2.430	1.653	0.709
Empathy: Perspective taking x Victim-N2 outgroup vs ingroup	0.482	1.152	-1.777	2.740	0.676
Empathy: Personal distress x Victim-N1 outgroup vs ingroup	-0.049	0.958	-1.926	1.828	0.959
Empathy: Personal distress x Victim-N2 outgroup vs ingroup	0.573	0.928	-1.245	2.391	0.537
Tobit log-scale	2.446	0.057	2.334	2.558	<0.001***

The H3 Victim model shows focal evidence for 2 hypothesis terms. Empathy: Fantasy is associated with higher predicted judgement (estimate = 2.04, p = 0.017*). Empathy: Perspective taking is associated with higher predicted judgement (estimate = 1.68, p = 0.087+). Session dummies remain in the fitted estimator for adjustment but are intentionally omitted from the substantive narrative.

14.6 H3 Bystander coefficient table

Table 21. H3 Bystander coefficient estimates

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Intercept	2.237	4.049	-5.698	10.173	0.581
Empathy: Fantasy	1.337	0.928	-0.482	3.155	0.150
Empathy: Empathic concern	-2.025	1.209	-4.395	0.344	0.094+
Empathy: Perspective taking	0.002	1.030	-2.017	2.022	0.998
Empathy: Personal distress	-1.516	0.910	-3.300	0.269	0.096+
Bystander-victim outgroup vs ingroup	-3.515	2.601	-8.612	1.583	0.177
Bystander-N1 outgroup vs ingroup	-1.442	2.871	-7.070	4.185	0.615
Bystander-N2 outgroup vs ingroup	-0.129	2.740	-5.499	5.241	0.963
Victim-N1 outgroup vs ingroup	-2.124	1.215	-4.505	0.256	0.080+
Victim-N2 outgroup vs ingroup	-2.317	1.216	-4.701	0.066	0.057+
N1/N2 same faculty vs different	0.826	0.820	-0.781	2.434	0.314
Age	0.353	0.103	0.152	0.554	<0.001***
Socioeconomic status	0.418	0.295	-0.160	0.996	0.157
Woman participant	-0.791	0.696	-2.155	0.574	0.256

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Participant faculty: Engineering vs Humanities	0.868	0.625	-0.357	2.093	0.165
Bystander-N1 outgroup vs ingroup x Bystander-N2 outgroup vs ingroup	-2.421	1.405	-5.174	0.331	0.085+
Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup	2.833	1.574	-0.252	5.918	0.072+
Empathy: Fantasy x Bystander-victim outgroup vs ingroup	-1.335	0.759	-2.823	0.152	0.078+
Empathy: Fantasy x Bystander-N1 outgroup vs ingroup	-0.947	0.780	-2.476	0.582	0.225
Empathy: Fantasy x Bystander-N2 outgroup vs ingroup	0.016	0.856	-1.663	1.694	0.985

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Empathy: Empathic concern x Bystander-victim outgroup vs ingroup	1.081	1.084	-1.043	3.206	0.318
Empathy: Empathic concern x Bystander-N1 outgroup vs ingroup	1.081	1.022	-0.922	3.085	0.290
Empathy: Empathic concern x Bystander-N2 outgroup vs ingroup	-0.073	1.135	-2.298	2.152	0.949
Empathy: Perspective taking x Bystander-victim outgroup vs ingroup	0.142	0.900	-1.623	1.906	0.875
Empathy: Perspective taking x Bystander-N1 outgroup vs ingroup	0.355	1.034	-1.672	2.381	0.732

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Empathy: Perspective taking x Bystander-N2 outgroup vs ingroup	0.348	0.960	-1.534	2.230	0.717
Empathy: Personal distress x Bystander-victim outgroup vs ingroup	1.647	0.770	0.138	3.155	0.032*
Empathy: Personal distress x Bystander-N1 outgroup vs ingroup	0.501	0.827	-1.120	2.122	0.545
Empathy: Personal distress x Bystander-N2 outgroup vs ingroup	0.948	0.788	-0.597	2.493	0.229
Tobit log-scale	2.322	0.057	2.210	2.433	<0.001***

The H3 Bystander model shows focal evidence for 8 hypothesis terms. Empathy: Personal distress x Bystander-victim outgroup vs ingroup is associated with higher predicted judgement (estimate = 1.65, p = 0.032*). Victim-N2 outgroup vs ingroup is associated with lower predicted judgement (estimate = -2.32, p = 0.057+). Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup is associated with higher predicted judgement (estimate = 2.83, p = 0.072+). Session dummies remain in the fitted estimator for adjustment but are intentionally omitted from the substantive narrative.

H4 reminder: legacy `accept_target` corresponds to active `decision_target`, and legacy `accept_other` corresponds to active `decision_other`; both refer to row-dynamic `target/other` roles, not fixed N1/N2 identities.

14.7 H4 Victim coefficient table

Table 22. H4 Victim coefficient estimates

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Intercept	9.771	2.575	4.723	14.819	<0.001***
Target accepted	-16.984	1.072	-19.086	-14.882	<0.001***
Other negotiator accepted	-3.951	0.705	-5.332	-2.570	<0.001***
Age	0.042	0.102	-0.158	0.242	0.681
Socioeconomic status	0.284	0.273	-0.251	0.819	0.298
Woman partici-pant	0.464	0.548	-0.611	1.539	0.398
Participant faculty: Engineer-ing vs Humani-ties	1.364	0.595	0.197	2.531	0.022*
Target accepted x Other accepted	4.558	0.772	3.044	6.071	<0.001***
Tobit log-scale	2.032	0.051	1.932	2.132	<0.001***

The H4 Victim model shows focal evidence for 3 hypothesis terms. Target accepted is associated with lower predicted judgement (estimate = -16.98, p = <0.001***). Target accepted x Other accepted is associated with higher predicted judgement (estimate = 4.56, p = <0.001***). Other negotiator accepted is associated with lower predicted judgement (estimate = -3.95, p = <0.001***). Session dummies remain in the fitted estimator for adjustment but are intentionally omitted from the substantive narrative.

14.8 H4 Bystander coefficient table

Table 23. H4 Bystander coefficient estimates

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Intercept	7.699	2.718	2.371	13.026	0.005**

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Target accepted	-15.354	0.967	-17.249	-13.459	<0.001***
Other negotiator accepted	-4.262	0.662	-5.559	-2.965	<0.001***
Age	0.164	0.094	-0.020	0.349	0.081+
Socioeconomic status	0.206	0.246	-0.276	0.689	0.402
Woman partici-pant	0.098	0.543	-0.967	1.164	0.856
Participant faculty: Engineer-ing vs Humani-ties	1.174	0.515	0.165	2.183	0.023*
Target accepted x Other accepted	4.849	0.769	3.342	6.357	<0.001***
Tobit log-scale	1.934	0.050	1.835	2.033	<0.001***

The H4 Bystander model shows focal evidence for 3 hypothesis terms. Target accepted is associated with lower predicted judgement (estimate = -15.35, p = <0.001***). Other negotiator accepted is associated with lower predicted judgement (estimate = -4.26, p = <0.001***). Target accepted x Other accepted is associated with higher predicted judgement (estimate = 4.85, p = <0.001***). Session dummies remain in the fitted estimator for adjustment but are intentionally omitted from the substantive narrative.

H5 reminder: `N1/N2` are reconstructed structural slots (not fixed aliases of `target/other`), `group_target/group_other` are legacy audit fields, and legacy `accept_target/accept_other` map to active `decision_target/decision_other` for row-dynamic roles.

14.9 H5 Victim coefficient table

Table 24. H5 Victim coefficient estimates

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Intercept	4.069	3.411	-2.617	10.755	0.233

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Empathy: Fantasy	1.469	0.656	0.184	2.755	0.025*
Empathy: Empathic concern	0.420	0.899	-1.342	2.183	0.640
Empathy: Perspective taking	1.168	0.766	-0.334	2.670	0.127
Empathy: Personal distress	-0.316	0.716	-1.719	1.087	0.659
Victim-N1 outgroup vs ingroup	1.766	1.806	-1.774	5.305	0.328
Victim-N2 outgroup vs ingroup	4.155	2.198	-0.153	8.462	0.059+
N1/N2 same faculty vs different	-0.299	0.574	-1.424	0.826	0.603
Target accepted	-16.910	1.071	-19.010	-14.810	<0.001***
Other negotiator accepted	-3.881	0.699	-5.251	-2.512	<0.001***
Age	0.016	0.100	-0.180	0.212	0.870
Socioeconomic status	0.219	0.267	-0.305	0.743	0.412
Woman participant	0.444	0.572	-0.677	1.564	0.438
Participant faculty: Engineering vs Humanities	1.495	0.610	0.299	2.690	0.014*

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup	-0.161	1.217	-2.547	2.224	0.894
Empathy: Fantasy x Victim-N1 outgroup vs ingroup	-0.826	0.468	-1.744	0.091	0.077+
Empathy: Fantasy x Victim-N2 outgroup vs ingroup	-0.632	0.578	-1.764	0.500	0.274
Empathy: Empathic concern x Victim-N1 outgroup vs ingroup	-0.045	0.684	-1.386	1.296	0.948
Empathy: Empathic concern x Victim-N2 outgroup vs ingroup	-0.784	0.849	-2.447	0.879	0.356
Empathy: Perspective taking x Victim-N1 outgroup vs ingroup	-0.058	0.596	-1.227	1.111	0.922
Empathy: Perspective taking x Victim-N2 outgroup vs ingroup	-0.276	0.742	-1.731	1.179	0.710

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Empathy: Personal distress x Victim-N1 outgroup vs ingroup	0.026	0.572	-1.095	1.147	0.964
Empathy: Personal distress x Victim-N2 outgroup vs ingroup	-0.139	0.541	-1.199	0.921	0.797
Target accepted x Other accepted	4.445	0.762	2.950	5.939	<0.001***
Tobit log-scale	2.024	0.051	1.924	2.124	<0.001***

The H5 Victim model shows focal evidence for 6 hypothesis terms. Target accepted is associated with lower predicted judgement (estimate = -16.91, p = <0.001***). ***Target accepted x Other accepted is associated with higher predicted judgement (estimate = 4.44, p = <0.001***). Other negotiator accepted is associated with lower predicted judgement (estimate = -3.88, p = <0.001***). Session dummies remain in the fitted estimator for adjustment but are intentionally omitted from the substantive narrative.

14.10 H5 Bystander coefficient table

Table 25. H5 Bystander coefficient estimates

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Intercept	10.158	3.366	3.561	16.756	0.003**
Empathy: Fantasy	0.737	0.674	-0.583	2.058	0.274
Empathy: Empathic concern	-0.705	0.860	-2.391	0.981	0.412
Empathy: Perspective taking	-0.680	0.731	-2.112	0.753	0.352

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Empathy: Personal distress	-0.178	0.699	-1.549	1.192	0.799
Bystander-victim outgroup vs ingroup	-1.983	1.951	-5.807	1.840	0.309
Bystander-N1 outgroup vs ingroup	-0.272	1.921	-4.037	3.494	0.887
Bystander-N2 outgroup vs ingroup	-1.351	1.831	-4.939	2.237	0.461
Victim-N1 outgroup vs ingroup	-1.543	0.774	-3.061	-0.026	0.046*
Victim-N2 outgroup vs ingroup	-1.589	0.759	-3.076	-0.103	0.036*
N1/N2 same faculty vs different	-0.105	0.564	-1.210	1.001	0.853
Target accepted	-15.384	0.962	-17.270	-13.498	<0.001***
Other negotiator accepted	-4.305	0.665	-5.609	-3.001	<0.001***
Age	0.184	0.097	-0.006	0.375	0.058+
Socioeconomic status	0.168	0.248	-0.317	0.654	0.497
Woman participant	0.040	0.557	-1.051	1.132	0.942
Participant faculty: Engineering vs Humanities	1.064	0.516	0.053	2.075	0.039*

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Bystander-N1 outgroup vs ingroup x Bystander-N2 outgroup vs ingroup	-1.307	1.011	-3.289	0.675	0.196
Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup	2.410	1.069	0.315	4.505	0.024*
Empathy: Fantasy x Bystander-victim outgroup vs ingroup	-1.143	0.507	-2.137	-0.149	0.024*
Empathy: Fantasy x Bystander-N1 outgroup vs ingroup	0.119	0.515	-0.890	1.127	0.818
Empathy: Fantasy x Bystander-N2 outgroup vs ingroup	-0.133	0.580	-1.270	1.004	0.818
Empathy: Empathic concern x Bystander-victim outgroup vs ingroup	0.698	0.812	-0.894	2.289	0.390

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Empathy: Empathic concern x Bystander-N1 outgroup vs ingroup	-0.033	0.666	-1.338	1.272	0.960
Empathy: Empathic concern x Bystander-N2 outgroup vs ingroup	-0.162	0.786	-1.703	1.378	0.836
Empathy: Perspective taking x Bystander-victim outgroup vs ingroup	0.649	0.618	-0.561	1.860	0.293
Empathy: Perspective taking x Bystander-N1 outgroup vs ingroup	0.542	0.662	-0.756	1.840	0.413
Empathy: Perspective taking x Bystander-N2 outgroup vs ingroup	0.714	0.682	-0.624	2.051	0.296

term_label	estimate	std_error	conf_low	conf_high	p_value_display
Empathy: Personal distress x Bystander-victim outgroup vs ingroup	0.605	0.608	-0.588	1.798	0.320
Empathy: Personal distress x Bystander-N1 outgroup vs ingroup	-0.191	0.567	-1.302	0.919	0.736
Empathy: Personal distress x Bystander-N2 outgroup vs ingroup	0.626	0.541	-0.434	1.685	0.247
Target accepted x Other accepted	4.891	0.791	3.340	6.442	<0.001***
Tobit log-scale	1.928	0.050	1.830	2.026	<0.001***

The H5 Bystander model shows focal evidence for 7 hypothesis terms. Target accepted is associated with lower predicted judgement (estimate = -15.38, p = <0.001***). Other negotiator accepted is associated with lower predicted judgement (estimate = -4.31, p = <0.001***). Target accepted x Other accepted is associated with higher predicted judgement (estimate = 4.89, p = <0.001***). Session dummies remain in the fitted estimator for adjustment but are intentionally omitted from the substantive narrative.

15 Compliance checklist

Table 26. Pipeline compliance checklist

criterion	status	evidence
a) uses judgement	YES	All formulas model judgement directly.
b) repeated structure by id	YES	All fitted Tobit models use participant-cluster robust standard errors through cluster = id.
c) session grouping	YES	The active Tobit branch uses factor(session) in every formula and documents that choice explicitly instead of claiming a random session intercept.
d) no double count introduced by the pipeline	YES	Imported rows = 4860; final analytical rows = 4860; duplicated source row numbers introduced by the pipeline = 0.
e) victim and bystander treated differently	YES	Role-specific formulas are estimated separately and H2/H3/H5 use different relational blocks for victim and bystander.
f) decision_target and decision_other included where required	YES	H4 and H5 both include decision_target, decision_other, and their interaction.
g) sociodemographics included in every hypothesis model	YES	Every H1-H5 formula retains age, ses, sex_female, and faculty_player_factor.

16 Corrections relative to outdated notes

The current production branch supersedes earlier notes that used `control_hidden` wording for negotiator code `0`, narrowed H3 to additive-only empathy terms, or implied that the main estimator used `(1|session)`. The repository now documents the implemented estimator and the authoritative role-specific design directly.

17 Limitations

The production branch does not fit a full multilevel Tobit with explicit random participant and session intercepts inside the same estimator. Sparse relational cells can produce rank-deficient design matrices, so some interaction contrasts are dropped automatically and reported as such. The dynamic figures visualize model-implied predictions from the saved primary Tobit fits and should be interpreted jointly with the coefficient tables rather than as standalone causal effects.

18 Discussion

The empathy results speak to a mechanism in which moral judgement is not only a response to outcomes but also to dispositional social sensitivity. When empathy slopes vary across ingroup and outgroup relations, the findings support the original theoretical expectation that empathic orientation is filtered through perceived social closeness rather than operating as a uniform moral amplifier.

The ingroup/outgroup results matter because the experiment embeds judgement in a relational structure with two negotiators and role-dependent social ties. Victim and bystander models are therefore not interchangeable. In the victim role, the central question is how the judged negotiator and the counterpart relate to the harmed person. In the bystander role, the participant is socially external to the harm event, so judgement can depend on a broader map that includes bystander-victim, bystander-negotiator, and victim-negotiator alignments.

The decision terms sharpen the moral interpretation of the target-focused outcome. `decision_target` captures what the judged negotiator did, `decision_other` captures the counterpart's choice, and their interaction tests whether the meaning of one decision changes when the other negotiator accepts or rejects. This is substantively important because moral evaluations of the target can respond both to individual action and to the joint negotiation outcome.

Practically, the results speak to negotiation ethics and third-party evaluation. If moral judgement shifts with empathy, faculty closeness, and joint decision patterns, then perceived fairness in harmful negotiations is shaped by both dispositional and relational context. That has implications for how observers assign blame, excuse strategic behavior, or infer responsibility from coordinated action.

Methodologically, the active estimator is a two-sided Tobit with participant-cluster robust standard errors and session fixed effects. This is an honest production choice for a bounded repeated-measures outcome because it preserves the Tobit structure for `judgement`, adjusts within-participant dependence through clustering by `id`, and controls for session-level shifts through `factor(session)`. At the same time, it is not equivalent to a fully mixed Tobit with random participant and session intercepts, so dependence is handled through robust inference plus fixed-effects adjustment rather than a full hierarchical likelihood.

The main limitations follow directly from that estimator choice and from the sparsity of some relational cells. The production branch does not estimate a full mixed Tobit, some interaction contrasts may be dropped in rank-deficient subsets, and any substantive reading should remain tied to the coefficient tables and model-implied figures rather than to isolated p-values. Future work should compare these production estimates against stable multilevel censored models, test alternative role-specific interaction sets, and examine whether the same theoretical patterns replicate under additional institutional or cultural contexts.

19 Final audit note

The project now faithfully reflects the authoritative design in its production branch: `judgement` is the outcome, the long file remains the single source, one row remains one real observation, role-specific group definitions are used, `decision_target` and `decision_other` are modeled where required, and repeated measurements are handled through participant-cluster robust inference with `factor(session)`.

No rank-deficiency warning remained in the saved fit summary for this run.

Estimator limitation: the production estimator is still a two-sided Tobit with `factor(session)` and participant-cluster robust standard errors by `id`, not a full mixed-effects Tobit with random participant and session intercepts.

20 Technical appendix: predictor code map

Table 27. Predictor-to-code map (technical appendix)

Predictor	Code
iri_fs	FS
iri_ec	EC
iri_pt	PT
iri_pd	PD
victim_N1_groupingroup	V-N1 In
victim_N1_groupoutgroup	V-N1 Out
victim_N2_groupingroup	V-N2 In
victim_N2_groupoutgroup	V-N2 Out
bystander_victim_groupoutgroup	B-V Out
bystander_N1_groupingroup	B-N1 In
bystander_N1_groupoutgroup	B-N1 Out
bystander_N2_groupingroup	B-N2 In
bystander_N2_groupoutgroup	B-N2 Out

Predictor	Code
N1__N2__same__facultysame	SameFac
iri__fs:victim__N1__groupoutgroup	FS x V-N1 Out
iri__ec:victim__N2__groupoutgroup	EC x V-N2 Out
iri__pt:bystander__victim__groupoutgroup	PT x B-V Out
iri__pd:bystander__N1__groupoutgroup	PD x B-N1 Out
decision__target	Target Acc
decision__other	Other Acc
decision__target:decision__other	Target x Other
faculty__player__factorEngineering	Eng part.
sex__female	Woman
age	Age
ses	SES

21 Conclusion

The conclusions presented here should be interpreted as associational rather than causal. In this report, moral judgement was estimated with a two-sided Tobit model that handles censoring, repeated participant observations, and session adjustment through `factor(session)` with participant-cluster robust inference.

This conclusion section cross-references the full report structure. It should be read together with the H1-H5 equation summaries, the hypothesis significance summary by role, the full coefficient tables and interpretation summary, and the significance-driven figures.

21.1 Hypothesis-by-hypothesis synthesis

21.1.1 H1

H1 evaluates empathy as a direct predictor of moral judgement while retaining common controls. For cross-reference, see the H1 equation summary and the H1 role-specific coefficient tables and figures. In the victim role, the role-specific support summary highlighted: Empathy: Perspective taking**. In the bystander role, no focal term reached the $p < 0.10$ threshold in the role-specific summary table. Taken together, H1 suggests that the role position conditions how clearly empathy appears in the fitted judgement pattern.

21.1.2 H2

H2 reminder: `N1/N2` are structural slots reconstructed within each row, not fixed aliases of `target/other`; `group_target` and `group_other` are legacy source-audit fields, while active H2 models use reconstructed relational predictors.

H2 focuses on role-specific ingroup/outgroup structure. For cross-reference, see the H2 equation summary and the H2 role-specific coefficient tables and figures. In the victim role, no focal term reached the p < 0.10 threshold in the role-specific summary table. In the bystander role, the role-specific support summary highlighted: Bystander-N2 outgroup vs ingroup+; Victim-N1 outgroup vs ingroup+; Victim-N2 outgroup vs ingroup+; Bystander-N1 outgroup vs ingroup x Bystander-N2 outgroup vs ingroup+; Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup+. This comparison indicates that relational alignment cues are not equally informative across roles, and they can become more visible when participants evaluate as observers rather than as directly harmed actors.

21.1.3 H3

H3 combines empathy, group structure, and interaction terms. For cross-reference, see the H3 equation summary, the H3 coefficient tables, and the H3 interaction figures. In the victim role, the role-specific support summary highlighted: Empathy: Fantasy*; Empathy: Perspective taking+. In the bystander role, the role-specific support summary highlighted: Empathy: Empathic concern+; Empathy: Personal distress+; Victim-N1 outgroup vs ingroup+; Victim-N2 outgroup vs ingroup+; Bystander-N1 outgroup vs ingroup x Bystander-N2 outgroup vs ingroup+; Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup+; Empathy: Fantasy x Bystander-victim outgroup vs ingroup+; Empathy: Personal distress x Bystander-victim outgroup vs ingroup.* The H3 evidence should therefore be interpreted as a test of contextual moderation: empathy does not necessarily operate as a uniform slope when social distance changes.

21.1.4 H4

H4 reminder: legacy `accept_target` corresponds to active `decision_target`, and legacy `accept_other` corresponds to active `decision_other`; both refer to row-dynamic `target/other` roles, not fixed N1/N2 identities.

H4 tests decision terms directly through `decision_target`, `decision_other`, and their interaction. For cross-reference, see the H4 equation summary and the H4 role-specific coefficient tables and significance figures. In the victim role, the role-specific support summary highlighted: Target accepted***; Other negotiator accepted***; Target accepted x Other accepted***. In the bystander role, the role-specific support summary highlighted: Target accepted***; Other negotiator accepted***; Target accepted x Other accepted***. Across both roles, H4 is typically where the decisional mechanism is most clearly visible, because the target judgement is explicitly conditioned by joint negotiation outcomes.

21.1.5 H5

H5 reminder: `N1/N2` are reconstructed structural slots (not fixed aliases of `target/other`), `group_target/group_other` are legacy audit fields, and legacy `accept_target`/`accept_other` map to active `decision_target`/`decision_other` for row-dynamic roles.

H5 is the integrated specification combining empathy, relational terms, decisions, and interactions. For cross-reference, see the H5 equation summary, H5 coefficient tables, and H5 significance figures. In the victim role, the role-specific support summary highlighted: Empathy: Fantasy*; Victim-N2 outgroup vs ingroup+; Target accepted*; **Other negotiator accepted***; Empathy: Fantasy x Victim-N1 outgroup vs ingroup+; Target accepted x Other accepted***. In the bystander role, the role-specific support summary highlighted: Victim-N1 outgroup vs ingroup*****; Victim-N2 outgroup vs ingroup*****; Target accepted***; Other negotiator accepted*; **Victim-N1 outgroup vs ingroup x Victim-N2 outgroup vs ingroup*****; Empathy: Fantasy x Bystander-victim outgroup vs ingroup*****; Target accepted x Other accepted***. H5 should be read as a synthesis rather than a replacement of previous hypotheses: it shows how dispositional, relational, and decisional components coexist in one model.

21.2 Overall interpretation

Taken as a whole, the five hypotheses indicate that moral judgement in this experiment is multi-mechanistic rather than one-dimensional. Empathy-related terms can matter, relational alignment can matter, and decision terms can matter strongly, but their visibility changes by role and model context.

In practical terms, the report supports a role-contingent interpretation: victim-side judgement retains stronger dispositional content in some specifications, while bystander-side judgement often shows greater dependence on relational context, and both roles remain sensitive to the joint decisions made by negotiators.